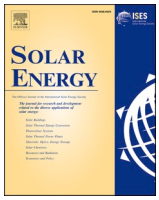
Solar Energy 292 (2025) 113445



Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener





A comprehensive review of radiative cooling technologies and their integration with Photovoltaic (PV) systems: Challenges, opportunities, and future directions

Kehinde Temitope Alao *, Syed Ihtsham Ul Haq Gilani, Kamaruzzaman Sopian, Taiwo Onaopemipo Alao, Zeshan Aslam

Mechanical Engineering Department, Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak Darul Radzuan, Malaysia

ARTICLE INFO

Keywords:
Radiative cooling
Photovoltaic systems
Passive thermal management
Energy efficiency
Material innovations

ABSTRACT

Effective heat management is crucial to enhance Photovoltaic (PV) performance and longevity. This review examines advancements in radiative cooling technologies, focusing on thermal coatings, photonic structures, and newly emerging materials with a dramatic improvement in PV efficiency. Compared to previous studies that focuses primarily on laboratory-scale development, this review takes the discussion further in the direction of large-scale feasibility, real-world application, and techno-economic considerations—concerns that are critical but yet to be discussed by previous reviews. It emphasizes optimization techniques enhanced by AI and adaptable cooling materials, offering a wider view on the integration of radiative cooling into next-generation PV systems such as tandem and perovskite solar cells. Furthermore, while most studies analyze individual cooling strategies in isolation, this review offers a comparative examination of various strategies, highlighting both advantages and disadvantages. Critical challenges such as scalability, environmental hardness, and cost-effectiveness—are critically analyzed, filling existing gaps in long-term performance analysis and field deployment strategies. By bridging the gaps between experimental studies and market-deployable implementations, this review is a valuable resource for enabling radiative cooling in renewable energy systems.

1. Introduction

Efficient thermal management is a critical factor for the sustainability of the energy conversion efficiency of Photovoltaic (PV) systems and ensuring their long-term reliability. Photovoltaic modules are regularly exposed to elevated ambient temperatures, which negatively impact their efficiency under sunlight [1]. A rise in temperature by 1 °C leads to a reduction of 0.4–0.5 % in the efficiency of silicon-based photovoltaic modules [2,3]. Extended periods of exposure to high temperatures reduce the lifespan of Photovoltaic (PV) modules by increasing material degradation [4]. To mitigate these challenges, researchers have extensively investigated numerous cooling techniques. In this context, radiative cooling emerges as an effective way of passive cooling [5,6]. Radiative cooling allows PV modules to disperse heat by emitting thermal radiation through the atmosphere's transparency window (8–13 μm), thereby counteracting overheating effects and enhancing efficiency. Fig. 1 presents a schematic of the interplay between solar irradiation, material emissivity, and the cooling power of selective and non-selective emitters.

This cooling technique depends on the capacity of materials to release infrared radiation to the cold upper atmosphere or space without requiring any external energy source. Therefore, radiative cooling reduces PV module temperatures, increases power generation, and decreases thermal stress, making solar power generation efficient and sustainable. Wei et al. [7] provides a comprehensive review of radiative cooling in solar cells based primarily on material development and night-time cooling integration. However, their discussion is lab-scale based with little focus on scalability, real-world implementation, and techno-economic considerations. This review broadens the scope with the inclusion of AI-optimized approaches, multimodal cooling materials, and mass deployment solutions, not addressed in previous reviews. In addition, we discuss the influence of radiative cooling on future photovoltaic technology, such as tandem and perovskite solar cells, with a focus on their commercialization and mass deployment.

Radiative cooling technologies have garnered increased attention as passive methods to combat thermal management challenges in

* Corresponding author.
E-mail address: kehinde_22012050@utp.edu.my (K.T. Alao).

https://doi.org/10.1016/j.solener.2025.113445
Received 20 January 2025; Received in revised form 9 March 2025; Accepted 18 March 2025
Available online 25 March 2025
0038-092X/© 2025 International Solar Energy Society. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Photovoltaic (PV) systems. Such technologies dissipate excess heat from PV modules via thermal radiation into the atmosphere or outer space, thereby reducing operational temperatures and mitigating efficiency losses [8,9]. Recent research has investigated innovative materials, coatings, and photonic structures to improve cooling efficiency in PV systems. Significant methods consist of wide-range emissive coatings that enhance infrared emissivity and simultaneously reflect solar radiation. For instance, Cui et al. [10] demonstrated that back surface-deposited coatings on PV tiles reduced temperatures by 3.6 °C and enhanced efficiency by 0.33 %, while Mandal et al. [11] investigated porous PVDF-HFP coatings and achieved a 6 °C cooling effect. These coatings are versatile since they can be deposited by spray or brushing, but long-term stability needs to be determined and investigated.

In addition to coatings, photonic structures have been investigated for their capability to potentially increase radiative cooling efficiency with light control. Raman et al. [12] designed a photonic radiator that cooled to 4.9 °C below ambient temperature. Although these structures yielded favorable results, their integration—particularly in SWCNT designs—is limited by high fabrication costs and poor scalability. On the other hand, Zhai et al. [13] addressed the scalability issue by developing a two-layer polymer-glass microsphere structure using a roll-to-roll method, achieving a similar cooling rate. However, the investigation was primarily based on the performance of materials rather than the actual application of PV modules. Further material innovations have contributed to the integration of radiative cooling technologies with PV systems. Zeng et al. [93] developed a new material called 'transparent radiative heat mirror' (TRHM) from cheap materials and applied this to perovskite solar cells where the temperature of the surface is reduced by 7.2 °C and the conversion efficiency is increased by 0.81 percent. While this approach shows great promise in developing cost-effective materials and methods, challenges related to stability and large-scale economic feasibility remain unresolved. Similarly Fathabadi [14] synthesized silica-based glass with nanopyramid coatings that lowered PV temperature by 9.2 °C under high irradiance conditions but not under different climatic conditions. Table 1 focuses on outlining several approaches, principal findings, and drawbacks of recent studies in the field of radiative cooling PV systems.

Although remarkable progress has been made in the radiative cooling of photovoltaic systems, there are some key areas that remain unaddressed and stand in the way of the application and efficiency of this concept. Scalability and fabrication costs are one of the most critical issues, meanwhile, as the system size increases, the probability of interconnect failure does not decrease. Various thermal management solutions, including photonic radiators and nanostructured coatings, provide a high level of cooling; nevertheless, complex manufacturing techniques and high costs exclude broad application. For example, photonic radiators which can provide a temperature drop of 4.9 °C below ambient temperature still pose challenges in fabrication. There is a need to overcome these barriers of scalability and cost to increase the utility of these technologies in lasting and effectively decreasing the cost of renewable energy so that investors may embrace the use of PV systems. The last serious drawback is that environmental testing conducted by most organizations is not broad. Many studies primarily assess cooling performance under controlled laboratory conditions, overlooking extreme scenarios such as high humidity, fluctuating temperatures, or particulate matter exposure. This lack of comprehensive testing frameworks poses a risk to the reliability and adaptability of radiative cooling technologies in real-world applications. This expansion will increase the reliability of the tests in various regions to make stakeholders confident, hence boosting commercialization.

Durability and long-term performance are other key considerations. Porous radiative coatings and nanostructured materials demonstrate

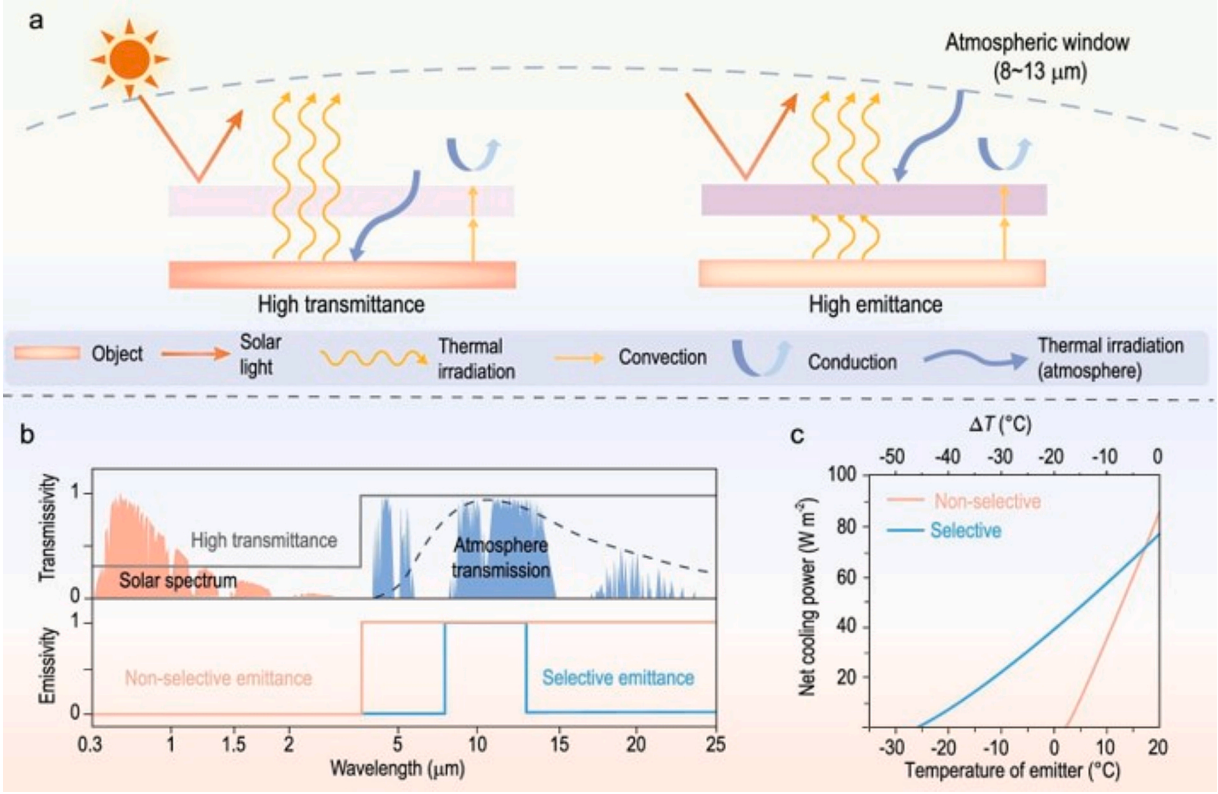


Fig. 1. Radiative cooling mechanism for photovoltaic (PV) systems. (a) Schematic representation of high transmittance and high emittance surfaces for passive heat shedding through the atmospheric window (8–13 μm). (b) Spectral transmittance and emissivity comparison showing the role of selective emittance in cooling performance. (c) Net cooling power vs. emitter temperature for selective and non-selective radiative cooling materials.
(Source: Adapted from [146])

Table 1
Summary of recent studies on radiative cooling techniques for photovoltaic (PV) systems.

Author	Study	Methodology	Key findings	Gaps or limitations	Ref.
Cui et al.	Thermal analysis of radiative cooling coating on the rear surface of photovoltaic tile	Proposed a thermal model and conducted simulations and outdoor experiments to assess PRC coatings' performance on PV tiles.	PRC coating with broadband emittance on the rear surface improved cooling by 3.6 °C and efficiency by ~ 0.33 %.	i) Confined to just two varieties, namely, broadband, and selective emissive coatings. ii) Focus on rear surface; front surface potential underexplored. iii) Assumptions such as uniform temperature and steady-state limit real-world applicability. iv) The environmental conditions were narrow, missing extreme scenarios. v) Lacks long-term performance and degradation data. vi) vi) No large-scale production or cost analysis.	[198]
Raman et al.	Photonic radiator for diurnal cooling	Fabricated a novel photonic radiator for outdoor use.	Achieved a 4.9 °C temperature reduction below ambient during daytime.	High costs and limited scalability due to complex fabrication processes.	[12]
Zhai et al.	Two-layer polymer-glass microsphere structure	Developed a scalable polymer matrix with embedded glass microspheres for radiative cooling.	Achieved sub-ambient cooling like advanced photonic structures using roll-to-roll manufacturing techniques.	Limited testing on real-world PV systems; focused on material properties.	[13]
Mandal et al.	Porous PVDF-HFP radiative cooling coating	Fabricated hierarchically porous coatings with high solar reflectance and infrared emittance.	Achieved 6 °C sub-ambient cooling under 890 W/m^2 solar irradiance; coating can be sprayed or brushed on substrates.	Mechanical durability and long-term performance under environmental stress have not been extensively evaluated.	[11]
Fathabadi	Silica-based glass with nanopyramid coating	Fabricated nanopyramid coatings for PV module cooling.	Reduced PV module temperature by 9.2 °C under 999 W/m^2 irradiance.	Real-world durability under varying climatic conditions has not been studied.	[14]
Lee et al.	Visibly transparent flexible radiative cooling coating	Designed and applied flexible coatings to crystalline silicon PV modules.	Decreased temperature by 7.7 °C on average; cooling reduced to 6.7 °C with higher humidity.	Limited to laboratory-scale studies; impact of prolonged exposure not evaluated.	[199]
Lu et al.	Silica-based coating with pyramidal texture	Developed ultra-broadband pyramidal-textured coatings for PV modules.	Improved conversion efficiency by 3.13 %.	Focused on front surface cooling; rear surface potential unexplored.	[200]
Cui et al.	Thermal and electrical performance analysis of monofacial double-glass photovoltaic module with radiative cooling coating on the rear surface.	Coupled thermal-electrical modeling, material optimization, and outdoor experiments to evaluate cooling and electrical efficiency of TPX/SiO2 coatings.	TPX/SiO_2 coating on the rear surface reduced PV module temperature by ~ 1 °C and increased efficiency by 0.21 %.	i) Focused on rear surface; front surface interaction unexplored. ii) Cooling effect is limited under real-world conditions (~1°C). iii) iii) Long-term performance and environmental durability remain unexamined.	[10]
Zeng et al.	Transparent-radiative heat mirror (TRHM) for cooling silicon PV modules.	Fabricated TRHM with ITO layer and honeycomb-like PDMS using a low-cost method. Conducted experiments to analyze thermal and electrical performance.	PV modules with TRHM achieved a 7.2 °C temperature reduction and a 0.81 % increase in power conversion efficiency (PCE).	i) Limited to a single combination of materials; other configurations unexplored. ii) Durability and performance under varying environmental conditions untested. iii) iii) Scale-up and manufacturing challenges not addressed.	[93]
Zhao et al.	Radiative cooling of solar cells with micro-grating photonic cooler.	Designed and fabricated a micro-grating silica cooler with enhanced emissivity and slight anti-reflection properties. Conducted outdoor experiments to assess performance.	The micro-grating silica cooler improved emissivity to ~ 0.9 and reduced commercial silicon cell temperature by 3.6 °C.	i) Cooling effect (3.6 °C) is moderate compared to other solutions. ii) ii) Focused on single configuration; scalability and cost not extensively addressed.	[215]
Zhou et al.	Superhydrophobic composite coatings for durable and efficient radiative cooling.	Combined FEP with modified Al_2O_3 particles using spray and phase separation methods to enhance hydrophobicity and optical properties.	Achieved 4.9 °C cooling under direct sunlight, radiative cooling power of 81.2 W/m^2, and excellent durability in extreme conditions (pH 1 to 14, UV exposure).	i) Limited scalability due to complex preparation methods. ii) Focus on building applications; adaptation for PV modules requires further investigation. iii) iii) Long-term degradation under real-world conditions not fully explored.	[216]

excellent short-term performance at high temperatures; however, their durability against prolonged exposure to atmospheric stressors (including UV radiation, temperature cycles, and chemical pollutants) is less well studied. This leads to higher maintenance costs over time and decreased reliability of systems because of the lack of transfer of knowledge. Deriving better material durability leading to lower operational costs will increase the adoption of these solutions if they meet the demands of the real-world conditions. Moreover, insufficient techno-economic evaluations have been a major bottleneck for radiative cooling technologies from transition from lab and research prototypes to market-ready solutions. Innovative designs suggest that PV efficiency can be improved, but their production feasibility, cost-effectiveness and ROI are often not considered. These evaluations are critical for fostering partnerships between academia, industry, and policymakers and encouraging investment in scaling these technologies.

Additionally, the combination of front and rear cooling techniques is not well-studied. Many studies focus merely on either the front or back surface, overlooking chances for synergistic effects that might improve thermal management and energy efficiency. Investigating integrated cooling techniques may improve efficiency gains, improve temperature control, and prolong the longevity of PV systems, thereby making solar energy production more sustainable. Finally, most of the current development into material innovations focuses simply on thermal regulation without the addition or exploration of adding other features such as self-cleaning or anti-soiling capabilities. These further characteristics are in great need and will help facilitate low maintenance and peak performance, especially within regions with pronounced dust or pollution levels. Creating multi-functional materials would greatly improve the performance and attractiveness of radiative cooling technologies, meeting the practical demands of PV systems in different situations. These challenges require interdisciplinary collaboration among material sciences, environmental engineering, and economic modeling. By overcoming scalability challenges, durability, adaptability in environmental conditions, and cost of the technology, radiative cooling technologies will be in a position to evolve from experimental ideas towards scalable solutions to drive the PV industry ahead and contribute positively to global renewable energy goals.

The review presented herein addresses some of the key gaps in the literature on radiative cooling technologies for PV systems. Most works done to date have enhanced specific cooling approaches considerably, with very few presenting broad-based discussions that could help narrow the gap between laboratory-scale developments and practical applications. This work aims to fill these gaps with the following novelty:

1. Comprehensive analysis across cooling strategies:

Most previous reviews have narrow limits to specific methodologies: coatings, photonic structures, or material innovations. This review compares all significant methods of radiative cooling with a synthesis of data from many different studies into commonalities and trade-offs among performance, scalability, and cost.

2. Practical emphasis on real-world applicability:

One of the unique features this review has is in the approach to real-world implementation with the perspective of radiative cooling technologies. The presented review reassesses various methods for performance against the widest range of climatic conditions by taking into account humidity, particulate matter, and temperature fluctuation-parameters that are very often neglected. This will set up a platform for forecasting the operational reliability of such technologies.

3. Emphasis on scalability and economic viability:

The lack of techno-economic studies concerning the different technologies for radiative cooling represents a lacuna in present research. The current review gives a broad techno-economic viability and commercial readiness for the different methods, while pointing to several challenges and opportunities in scaling up novel solutions to mass-market deployment. Insights from the work will be key to closing the gap from prototype towards market-ready products.

4. Integration with next-generation PV systems:

This review, for the first time, presents the radiative cooling technologies tailored for advanced PV systems like tandem and perovskite solar cells. The present study discusses how the advanced materials and technologies interact, pointing to future research toward better efficiency and durability of next-generation PV systems.

5. Long-term performance and multi-functional solutions:

Most existing studies neglect the long-term durability of radiative cooling solutions, which can undermine their practical relevance. This review highlights that the extensive testing under realistic conditions is necessary for the evaluation of long-term performance. Then, some multi-functional cooling materials with features such as self-cleaning, anti-soiling, and anti-reflection properties are discussed, showing how these innovations may contribute to better operational efficiency in PV systems.

6. Effect on the renewable energy sector:

Addressing thermal challenges in PV systems through innovative cooling solutions can significantly reduce the Levelized Cost of Electricity (LCOE) from solar power. The efficiency and durability of solar energy systems will therefore directly impact reaching global sustainability goals and transitioning toward renewable energy-based energy production.

These insights form the bedrock for collaborative innovation reviewed, with synopses of advanced developments across subjects including material sciences, environmental engineering, and economic modeling. The review sets the tone, therefore, by fusing the two concepts into one useful multidisciplinary reference with clear suggestions for researchers, industrial stakeholders, and policy-makers who look forward to actionable proposals toward advancing experimental radiative cooling concepts into real-world, large-scale sustainable products for the solar energy industry.

2. Advancements and trends in radiative cooling technologies

2.1. Overview of PV cooling techniques

Photovoltaic (PV) cooling systems have significantly developed to mitigate the thermal management issues of concern to efficiency, especially under conditions of hot climates. The additional heat buildup degrades PV conversion efficiency and promotes material degradation, calling for sophisticated cooling mechanisms. These cooling mechanisms can be broadly classified into passive, active, and hybrid cooling strategies with specific advantages and trade-offs.

2.1.1. Passive cooling techniques

Passive cooling techniques move heat without using external energy input, hence are inexpensive and environmentally friendly. Radiative cooling releases thermal radiation in the atmospheric window of transparency (8–13 μm), allowing for passive heat dissipation below ambient temperatures. Innovations like porous PVDF-HFP coating have achieved up to 6 °C temperature decrease, although scalability and long-term reliability remain issues [15,16]. Photonic radiators and nanostructures enhance emissivity, reducing heat buildup by 4.9–9.2 °C, but expensive production limits their application at large scale [17]. Transparent mirrors deliver up to 7.2 °C cooling and a 0.81 % efficiency

improvement, with scalability advantages but moderate durability concerns [215]. The performance of these passive cooling techniques is illustrated in Fig. 2(a), which compares their temperature reduction capabilities.

Phase change materials (PCMs) employ latent heat storage to capture surplus thermal energy, reducing PV panel temperatures by more than 10 °C and enhancing efficiency by as much as 6 % [18,19]. Air cooling mechanisms, including natural convection, fans, and finned heat sinks, can lower temperatures by as much as 10 °C, although their efficiency is affected by surrounding conditions [20,21]. The effect of passive cooling techniques on efficiency can also be observed from Fig. 2(b), where improvement in efficiency is plotted for the various techniques. A key factor in the choice of a passive cooling technique is the compromise between scalability and cost. Fig. 2(c) summarizes such compromises, describing how photonic radiators provide high-performance cooling at high cost, while air cooling and radiative cooling are still more cost-effective and scalable [20,21] The time-dependent performance of some of the chosen passive cooling methods is also presented in Fig. 2 (d), describing differences in cooling efficiency during the day [22,23].

2.1.2. *Active cooling techniques*

The performance and longevity of PV modules depend highly on the cooling techniques to prevent excessive module temperature. Main techniques are water cooling, forced air cooling, TEC, and liquid immersion cooling, and all these range from being very effective to practical in large arrays. Water cooling uses circulating water over or at the back of PV modules. This method reduces temperatures from 76.8 °C to 70.1 °C, thus increasing efficiency from 8 % to 9.8 % [24]. It was shown through a study that front and back water-cooling combined enhanced output power by 20.9 % and efficiency by 52.7 % in summer [25]. The change in output power in relation to cooling flow rate is depicted in Fig. 3(a), while the resulting improvements in efficiency are illustrated in Fig. 3(b). Water cooling systems, however, require energy for pumping, which increases overall power consumption and operational expenses. The impact of varying cooling flow rates on temperature rise is plotted in Fig. 3(c), with higher water flow rates relating to reduced PV module temperatures. Likewise, degradation of electrical efficiency with time for various cooling rates is plotted in Fig. 3(d), wherein cooling reduces efficiency loss as compared to no cooling.

Forced air cooling enhances PV heat dissipation, reducing temperatures and efficiency by 7–10 °C and 0.67–0.80 %, respectively [26]. Its power consumption is, however, too high to offer feasibility for use in warm climates. Thermoelectric cooling (TEC) is efficient but consumes power, decreasing PV temperatures by 21 °C (Fig. 4a) and enhancing efficiency by 11.23 % (Fig. 4b) [27]. TEC enhances energy output by a further 8.3 %, with higher power consumption (Fig. 4d). The temperature-efficiency plot (Fig. 4c) supports TEC's consistency at elevated efficiency. With phase change materials (PCM), TEC controls module temperatures and enhances efficiency by a further 2.5–3.5 % [28]. Despite its benefits, TEC is costly and energy-intensive, and therefore limited to small-scale implementation.

Liquid immersion cooling efficiently dissipates heat in PV modules, showing promise in concentrated PV systems by significantly lowering temperatures and improving efficiency [29,30]. However, its expensive nature, sophistication, and materials degradation restrict broad application. Active cooling techniques, like the utilization of fans and water,

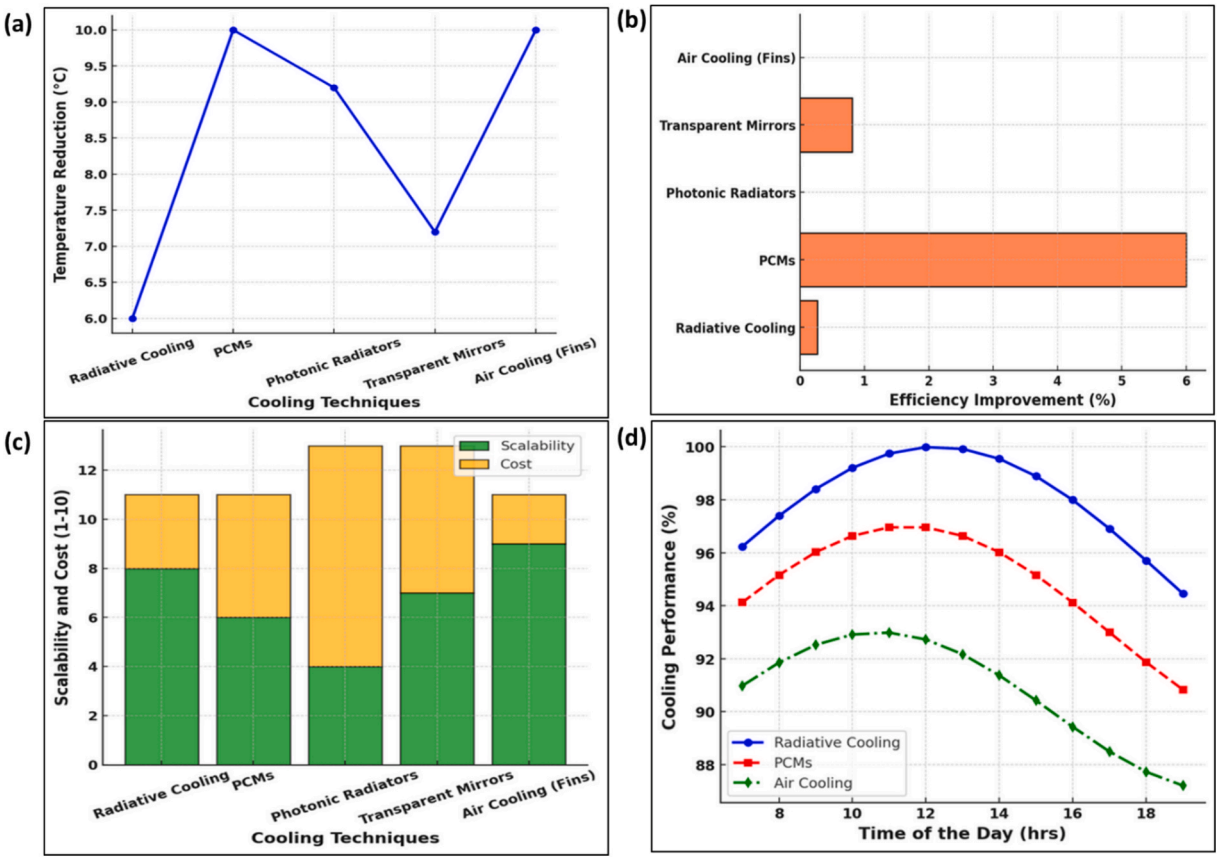


Fig. 2. Comparative analysis of passive cooling techniques (a) Temperature reduction achieved by passive cooling methods, including radiative cooling, PCMs, photonic radiators, transparent mirrors, and air-cooling fins (Adapted from [15,16,17]) (b) Efficiency improvement of passive cooling techniques, with PCMs providing the highest enhancement, followed by transparent mirrors and radiative cooling (Adapted from [19,129]) (c) Scalability vs. cost trade-offs of passive cooling methods, where photonic radiators have high efficiency but higher costs, while radiative and air cooling offer better scalability (Sourced from [20,21]) (d) Time-dependent cooling performance of radiative cooling, PCMs, and air cooling, showing efficiency variations throughout the day (Reconstructed from [22,23]).

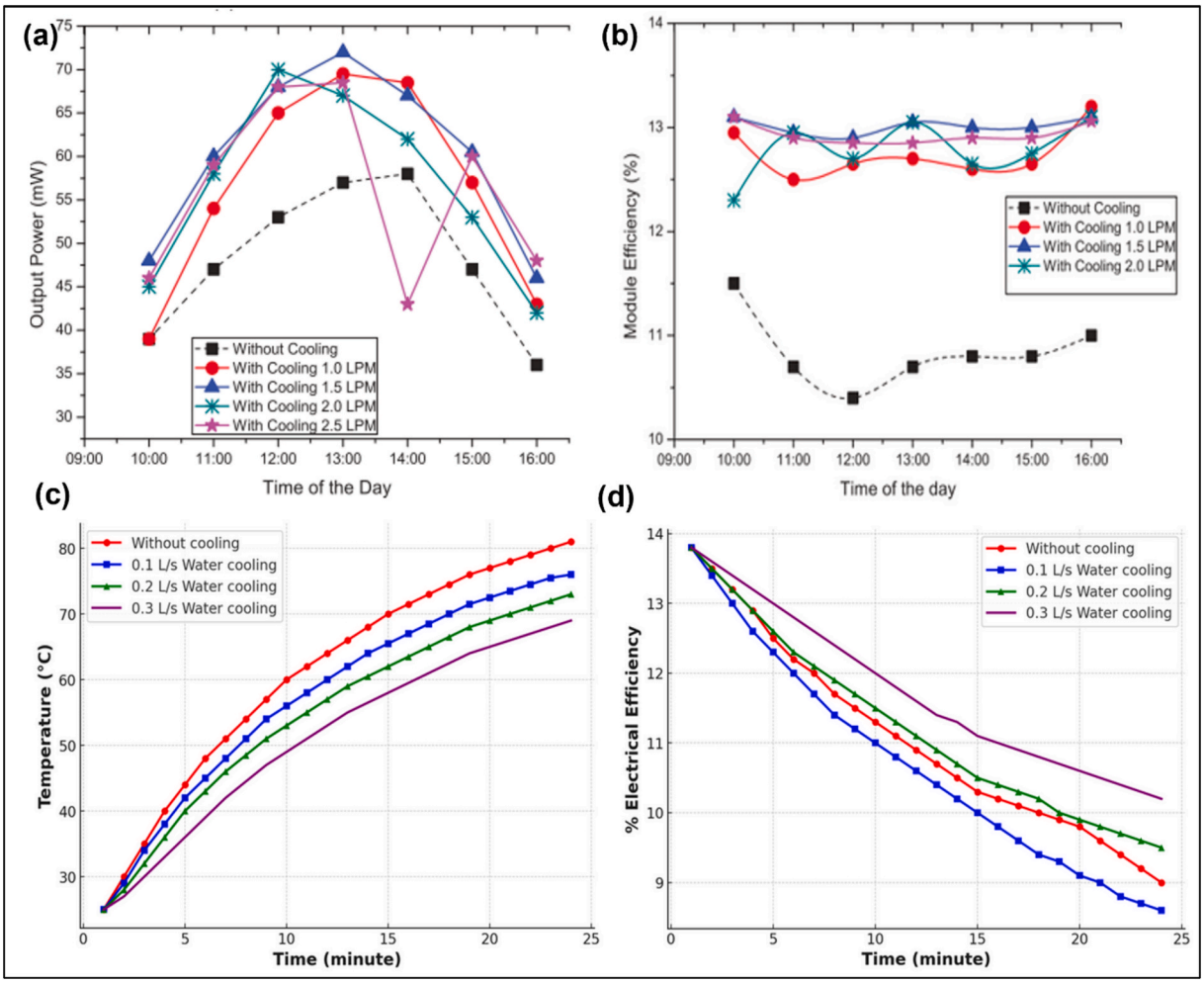


Fig. 3. Effect of water cooling on PV module performance. (a) Output power variation with and without water cooling at different flow rates (b) Efficiency improvement of PV modules under different cooling flow rates compared to no cooling (c) Temperature rise of PV modules under various water-cooling flow rates and without cooling (d) Electrical efficiency reduction over time under different water-cooling flow rates and without cooling
(Source: Adapted from [25][24]).

enhance Photovoltaic (PV) efficiency but increase energy consumption and maintenance costs, leading to a Levelized Cost of Electricity (LCOE) of 0.096 to 0.159 €/kWh [31,32,33]. These constraints confine active cooling to be impractical in off-grid locations [34]. Among active cooling methods, water-based cooling with microchannel heat sinks is the most scalable, offering high temperature reduction and efficiency improvement and improving temperature uniformity [34,35]. Water and TEC cooling offer the highest efficiency gains, though TEC remains costly and energy-intensive. Forced air cooling is less efficient but simpler to apply with very low temperatures, and liquid immersion cooling offers promise in specialized markets. Microchannel water cooling offers a promising solution for commercial PV installations, but operational and economic issues exist.

2.1.3. Hybrid cooling techniques

Hybrid cooling techniques are the combination of passive and active methods to improve Photovoltaic (PV) systems' thermal management, minimizing efficiency losses due to elevated temperatures. Experimental results have demonstrated that hybrid cooling lowers PV module temperatures substantially over conventional single-mode approaches. For instance, a PV/TEG nanofluid cooling lowered temperatures as much as 69.55 °C, whereas PCM with water cooling lowered temperatures by 12.6 °C [36,37]. Further, hybrid fin cooling and PCM realized a temperature decrease of 32 °C, while TEG through Ag-H_2O nanofluid cooled temperatures by 13.17 °C [38,39]. The temperature decreases have the cost of higher performance, especially at summer peak and high irradiation conditions when hybrid cooling always outshines traditional cooling methods.

Beyond temperature reduction, hybrid cooling techniques contribute to efficiency improvements and energy output gains. The integration of water and nanofluids increased PV efficiency by 20.68 %, while hybrid cooling with fins and water flow improved efficiency from 11.5 % to 13.3 %, with corresponding power output gains of up to 7.92 % [38,40]. Moreover, thermoelectric-PCM hybrid systems achieved a 19.4 % improvement in efficiency, and PV/T hybrid systems with PCM, forced convection, and nanofluids reported efficiencies of up to 85 % [41,42]. These developments make hybrid cooling a feasible option for enhancing the lifespan of PV systems and maximizing power output under varied climatic conditions.

However, application in practice of hybrid cooling is complicated by issues of energy consumption, cost, and uncertainty in field performance. Hybrid adsorption-compression cooling systems, despite a 62.5 % reduction in electricity usage, come with a payback period of 9.65 years, and thermoelectric cooling systems are still high on energy usage [43,44]. Moreover, laboratory-tested cooling technologies are expected to obtain greater efficiency gains compared to outdoor applications because environmental conditions like solar radiation, wind speed, and humidity vary randomly. For example, water-film hybrid cooling

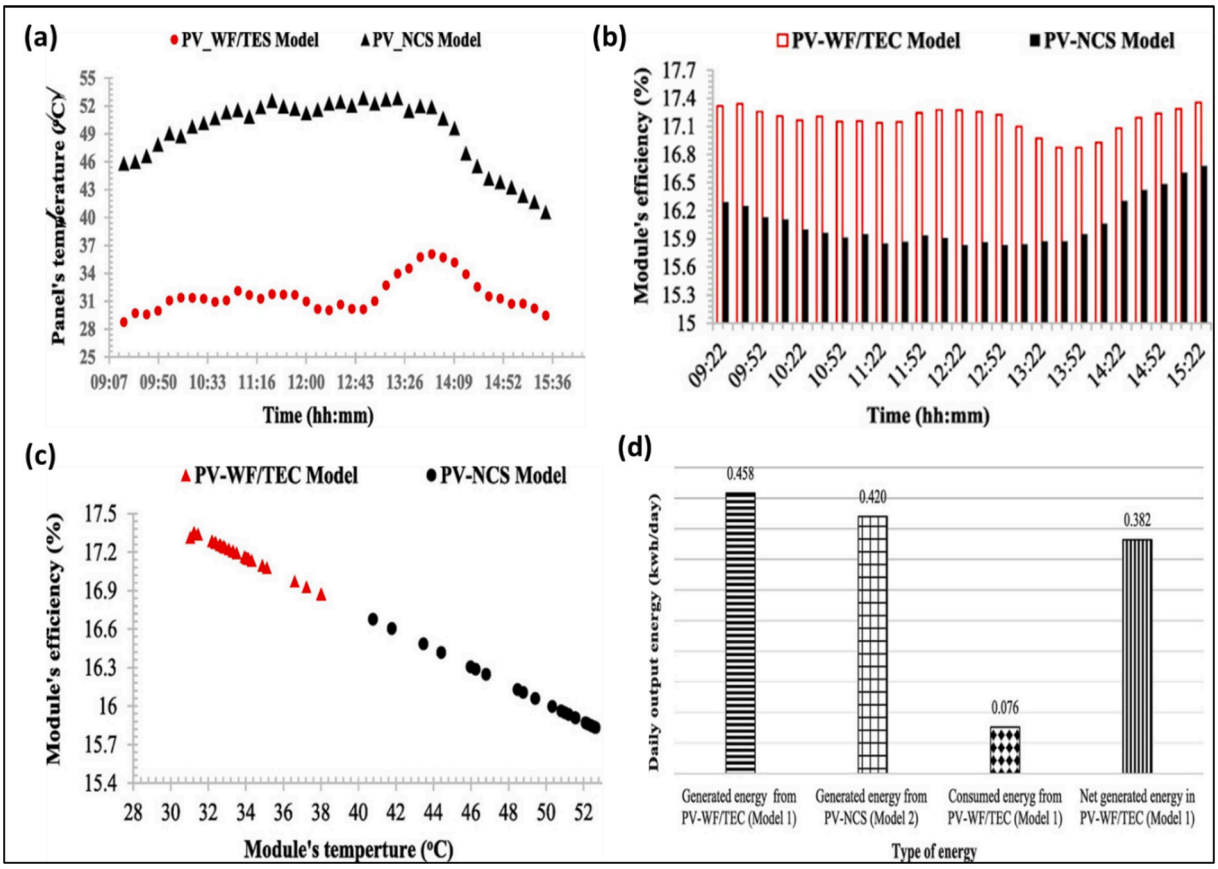


Fig. 4. TEC performance analysis (a) TEC-cooling-induced lowering of PV module temperature compared with non-cooled systems (b) Efficiency gain of PV modules under TEC cooling (c) Module temperature vs. efficiency in non-cooled and TEC-cooled systems (d) TEC system and conventional PV module daily output energy and net energy generation. Copyright 2024, Elsevier (License Number: 5981911084697).
(Source: Adapted from [27])

systems lowered the PV temperature by 15–19 % in actual applications compared to up to 35 % in laboratory experiments [45]. Despite these limitations, hybrid cooling continues to demonstrate economic and technical feasibility, particularly for large-scale solar power plants where energy savings and extended PV lifespan justify initial investments.

Compared to active and hybrid cooling technologies, radiative cooling has the special benefit of being capable of dissipating heat passively without drawing power from an external power source and hence presents a low-cost and environmentally friendly solution. Its cooling performance is typically lower than that of active technologies such as water cooling or TEC, which have a larger temperature difference available but consume high levels of energy input. Radiative cooling is also subject to atmospheric transparency and performs less well in polluted or humid atmospheres. In contrast, hybrid cooling systems—such as PCM-TEC or nanofluid-water cooling—combine passive and active techniques to achieve greater efficiency gains, but at a higher operational cost. While radiative cooling is ideal for low-power, maintenance-free PV applications, active and hybrid methods remain superior in high-irradiance conditions where aggressive thermal management is required.

2.2. Comparative analysis of cooling methods

2.2.1. Thermal coatings

Cooling technologies for Photovoltaic (PV) systems have been advancing at a rapid rate, mainly in response to addressing heat-induced inefficiencies, particularly in warm climatic conditions. These advancements enhance PV efficiency through new radiative, passive, and active cooling technologies that aim at temperature-induced inefficiencies. Radiative cooling coatings are among the most effective approaches. Nanophotonic coatings, for instance, enhance cooling by increasing thermal emission in the infrared atmospheric transparency range, significantly lowering PV module temperatures [46]. Multifunctional coatings combine radiative cooling and sub-bandgap reflection, improving efficiency by lowering heat generation [204]. Coatings like Polyethylene Terephthalate (PET) and polydimethylsiloxane (PDMS) boost glass transmission within the solar region but lack low thermal radiance factor [204]. These features are translated into large reductions in surface temperature and improvement in overall cooling efficiency. Fig. 5 demonstrates these mechanisms by a schematic of cooling methods and a comparative study of absorptivity and emissivity spectra of various Photovoltaic (PV) materials. Such effects are achieved in the form of large-scale surface temperature reductions and enhanced overall cooling efficiency.

Beyond material composition, the trade-offs regarding cooling efficiency must be taken into consideration when choosing the radiative coatings. Nanophotonic materials provide better cooling effectiveness but at a high cost compared to common PDMS or PET PV coatings [47]. Furthermore, some PV coatings such as silver-based metallic PV coatings present high reflectivity levels, efficient for cooling but interfering with visible light absorption. This creates the need for an optimization balance [48]. Another important aspect is durability in time—the high-emissivity PV coatings under UV can start to wear out faster than expected, raising concerns about the long-term performance of the PV system [49]. These trade-offs demonstrate that when selecting the potential coatings for the aforementioned purpose, it is important to choose the one that provides an appropriate balance between cooling

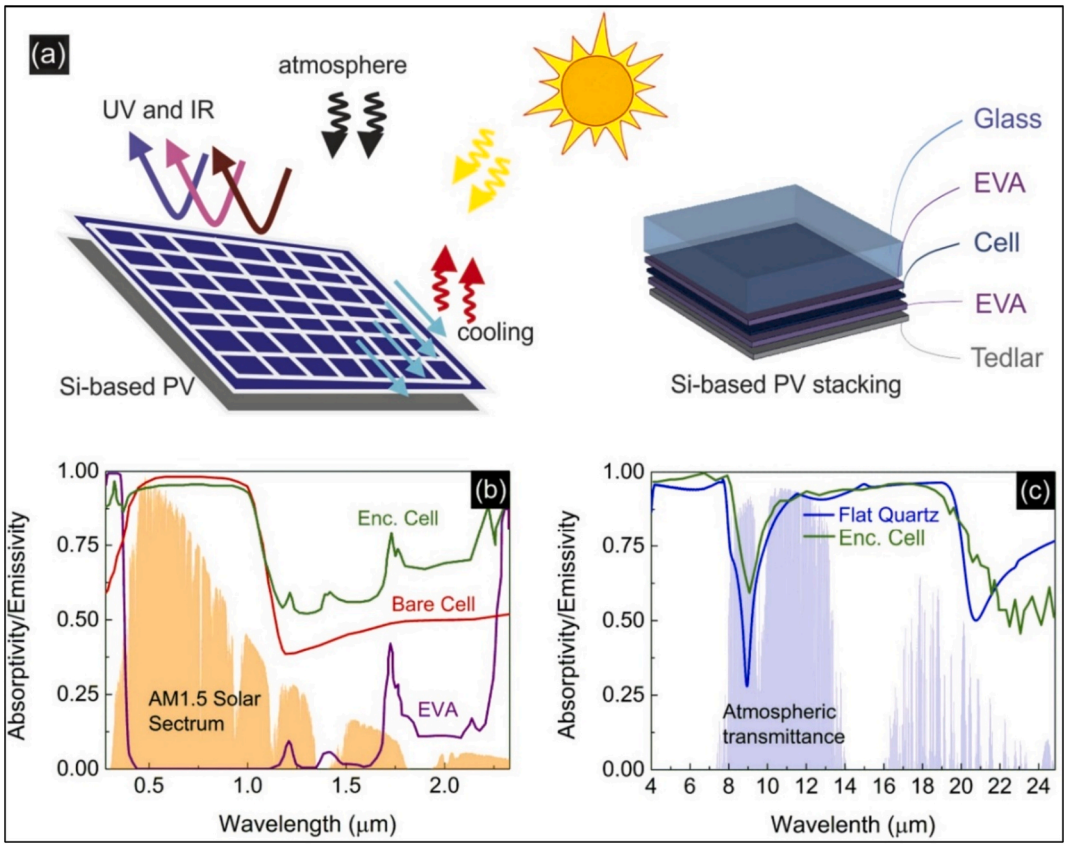


Fig. 5. (a) Schematic of radiative thermal management approaches for PV systems, including the material stacking of encapsulated crystalline silicon-based PVs. The diagram shows how different layers, such as glass, EVA, and Tedlar, function in thermal management. (b) Comparison of the absorptivity of a bare cell (red), an encapsulated cell (green), and an EVA wafer (purple), with improvements at each stage of the spectral performance. (c) Emissivity spectra in the mid-IR range for a fused quartz glass layer (blue) compared with the encapsulated cell (green), showing higher efficiency of radiative cooling (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
(Adapted from [46]).

efficiency, cost-efficiency, and stability in time.

Other thermal management strategies are provided by metallic and dielectric coatings. Metallic coatings like silver, while of high reflectivity, are commonly used but may absorb visible and near-infrared light. This can be overcome by using high index-low index periodic layers since they improve the spectral selectivity as shown in Fig. 6 [50]. Dielectric coatings such as aluminum oxide (Al_2O_3) can absorb selectively in particular spectral windows without compromising the visible and near-infrared absorption, thus maintaining a great cooling power [5]. The nano-coatings further extend the possibility for thermal management in PV systems. Further possibilities provided by nano-coatings allow thermal management in PV systems. Nano-titanium dioxide (TiO_2), nano-zinc oxide (ZnO), and nano-tin oxide (SnO) are the materials with exceptional hydrophobic features with low thermal insulation. These coatings help to minimize surface temperatures while also providing an extra protective barrier to environmental impacts, ensuring no depletion of their long-term lifetime [51].

Real-world applications of radiative cooling technologies have been

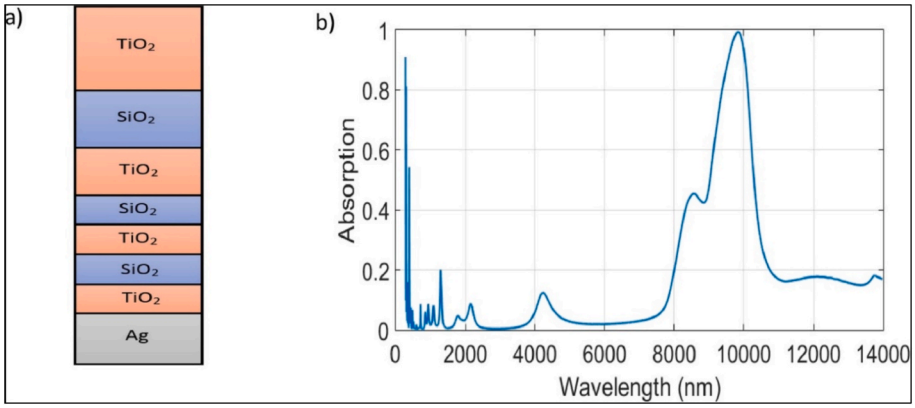


Fig. 6. (a) Schematic of the high index-low index periodic layer structure made of alternative TiO_2 and SiO_2 layers on top of a metallic silver substrate for enhanced spectral selectivity; (b) absorption spectrum showing the optical performance of this structure at different wavelengths (Adapted from [50]). Copyright 2017, Elsevier (License Number: 5981920835693).

thoroughly researched, especially in hot climates. Pilot projects initiated in arid and desert climates utilizing radiative PV coatings have proven to be effective as it has been found that radiative PV coatings allow for 5–10 % reduction in PV operational temperature, which leads to quantified improvements in PV energy yield [52,53]. On the other hand, urban and rural settings offer different challenges. Urban installations typically experience more dust accumulation, causing PV coatings to degrade in performance over a shorter time. Alternatively, rural PV units could offer more open-sky opportunities for radiative PV cooling [54]. These findings indicate the need to choose coatings according to environmental features to enhance the performance of PV.

Durability remains primary concern when choosing the radiative cooling coatings. Not all PV cooling materials are expected to last the same amount of time under specific conditions of increased UV exposure, mechanical strain, and humidity, which may outweigh their efficiency in the long run [55]. For instance, PV coatings made from polymers such as PET and PDMS are estimated to lose 5–15 % emissivity over the course of 5–10 years as a result of environmental wear and tear, requiring their replacement or cleaning in regular intervals [56,57]. Various studies have aimed at using hydrophobic nano-layers such as fluorinated polymers, nano-SiO_2 as a way of avoiding absorption of water to prevent dirt accumulation and maintain a clean surface, and thereby solving such problems and evaluating how such use positively influences overall PV system performance [58]. These temperature savings directly mean the enhancement of efficiency. For instance, radiative cooling combined with sub-bandgap reflection has the potential to significantly enhance PV efficiency [46], while front-surface water-spray cooling of Photovoltaic (PV) modules can result in up to 3.9 % improved PV efficiency [205].

The effectiveness of the thermal coatings is clearly shown through their ability to effect prominent reductions in temperature. Radiative cooling coatings can reduce PV module temperatures by up to 2.0 °C [211]. PET and PDMS coatings yield reductions of around 1.15 °C and 1.35 °C, respectively [204]. Further, advanced designs with low-iron soda-lime glass having porous and diamond layers showed even better potential and enabled temperature reductions by up to 14 °C compared with traditional coolers [59]. These temperature savings directly mean the enhancement of efficiency. For instance, radiative cooling combined with sub-bandgap reflection has the potential to significantly enhance PV efficiency [46], while front-surface water-spray cooling of Photovoltaic (PV) modules can result in up to 3.9 % improved PV efficiency [205].

Despite their advantages, there are several challenges that thermal coatings have to meet. The most important one is related to cost-effectiveness, especially in the case of multifunctional coatings combining multiple cooling strategies [211]. Material selection is also very important because the structural and optical properties of the coatings directly influence their performance. Future studies should concentrate on optimizing these characteristics to create new and better options for managing thermal energy in the case of PV systems [60]. However, thermal coatings have become a key component for current measures of PV cooling. Optimizing the temperature at the working surface and improving the efficiency increases the life expectancy of PV modules and the global objective of creating efficient and sustainable systems for world-wide solar energy production. Further innovation in material science and coating design is needed to overcome the challenges and fully realize the potential of the technology.

2.2.2. Photonic structures

Photonic structures are one of the key innovations in the area of cooling techniques that use unique physical mechanisms to realize thermal regulation and energy efficiency. Radiative cooling, in particular, passively emits thermal radiation within the atmospheric window (8–13 μm), cooling surfaces below ambient temperatures without energy consumption [61,62]. The selected process provides an energy-saving and sustainable solution to traditional cooling technologies. The advanced photonic materials investigated include silica-based photonic crystals, metal-dielectric nanostructures, and polymeric bio-inspired photonic films for PV-integrated cooling. These materials have a significant contribution to improving the PV performance through the improvement of radiative heat dissipation, a decline in overheating effects, and higher energy conversion efficiency [63,64].

Photonic coatings of PV modules were demonstrated to exert remarkable cooling and efficiency-enhancing impacts. Silica photonic crystals have more than 0.9 emissivity in the infrared range and exhibit very high transparency in the visible regime. Photonic coatings lower PV module temperature by 6–8 °C and correspondingly improve the energy conversion efficiency by 2–4 % [65]. Heat dissipation is augmented by metal-dielectric nanostructures, but with low parasitic absorption loss. For real-life PV applications, the coatings lower the temperatures by as much as 10 °C and provide long-term stability [66]. Polymeric bio-inspired films, which are built from nanostructured silica and lightweight polymers, provide passive cooling without compromising flexibility and hydrophobicity, thus preventing dust deposition. Experimental studies show a 10 % increase in PV energy output in hot climates when these materials are applied [67].

For large-scale adoption, the photonic coatings have to be incorporated into the PV modules without affecting flexibility, strength, or efficiency. A few challenges in manufacturing are material adhesion and optical functionality versus mechanical stress. Laser Interference Lithography (LIL) and Nanoimprint Lithography (NIL) are methods for producing large-area coatings cost-effectively [68,69]. However, stability in flexible PV modules is an issue, since photonic coatings are required to preserve their optical properties under bending [70]. Additionally, cost-performance tradeoff plays a vital role in establishing the economic feasibility of the photonic cooling materials for application in PV systems since profitability is vulnerable to both reducing the production cost and enhancing efficiency [61].

Pilot tests have confirmed the efficacy of photonic coatings under actual PV conditions. Under desert climatic conditions, photonic cooling films-coated PV panels had up to 10 % higher energy retention during peak sun hours, significantly mitigating overheating and ensuing efficiency loss [71,72]. Despite promising research findings, the challenges of scale-up and costs associated with the process cannot be overlooked before the technology may be considered for widespread commercial use. Where high-performance coatings yield quantifiable efficiency gains, scale-up and cost are again top priorities concerns for photonic cooling adoption by the PV industry.

Photonic structures have vast potential to transform next-generation cooling technologies with high-performance and sustainable solutions for PV thermal management. Additional development of optimization techniques can improve the scalability and performance of photonic structures to complement—and even outperform—in certain instances, traditional cooling methods. Ongoing development of these types of materials can create more widespread commercial use, which can make PV systems more robust, efficient, and responsive to environmental condition change.

2.2.3. Innovative materials

Advanced materials form the base upon which technologies are developed to improve the performance, scalability, and environmental adaptability of radiative cooling applications. Table 2 summarizes the strengths and limitations of major cooling techniques, giving a variety of methods and their respective benefits. As illustrated in Fig. 7, radiative cooling materials have evolved from traditional passive cooling techniques to advanced photonic structures, polymer-based coatings, and thermochromic materials. These innovations have played a crucial role in reducing thermal stress and improving Power Conversion Efficiency (PCE) in Photovoltaic (PV) modules.

Silica nanoparticle-photonic thin films and multi-layer coatings are vital in lowering PV cell temperature and improving efficiency. These coatings reflect sub-bandgap light, reducing heat absorption and module

Table 2
Strengths and limitations of cooling techniques.

Technique	Strengths	Limitations
Photonic Materials	High efficiency, scalable manufacturing	Complex preparation, high costs
Static and Dynamic Materials	Consistent performance, adaptability to conditions	Need for further optimization
Polymer-Based Materials	High emissivity, reduced heat conduction and convection	Matching ideal spectral characteristics
Metamaterials and Nano-photonics	Significant cooling under direct sunlight, advanced designs	Long-term environmental adaptability testing
VO_2 Composites	Adaptive thermal management, tunable transition temperatures	Precise engineering required for optimal performance

efficiency. Anti-reflective coatings based on silica nanoparticles can contribute substantially to improving transmittance, further improving PCE [73–74]. Discrete multilayer anti-reflective coatings like SiO_2 nanohelix arrays reduce total reflection and maximize photon-to-electron conversion efficiency [75].

Current research on self-assembling photonic materials is also driving passive cooling by selectively reflecting heat while passing useful wavelengths. Such materials, which are constructed from monodisperse spherical particles, are able to produce sharp spectral transitions to enhance cooling without sacrificing energy absorption [76]. Composite coatings synthesized from sol-gel methods, such as zirconium dioxide (ZrO_2) and titania (TiO_2), enhance optical transmittance, mechanical strength, and hydrophilicity and therefore are perfectly suitable for PV integration [77]. Another potential solution utilizes thermochromic vanadium dioxide (VO_2) coatings with an SMT temperature of 68 °C. At temperatures below this, the coating is transparent to optimize solar absorption, but above it, it is able to reflect infrared radiation and avoid overheating [78,79,80]. Optimized VO_2 coatings can boost PV efficiency by as much as 11 % and provide adaptive thermal management needed for long-term module stability [81].

Advancing material technologies are constantly revolutionizing radiative cooling technology. With time, technologies have advanced from passive cooling to intricate photonic coatings, polymer-based systems, and dynamic thermochromic materials. In addition to improving Photovoltaic (PV) efficiency, these advances contribute scalable and sustainable cooling technology. Even with substantial advancements, cost, bulk manufacturing, and environmental hardness remain challenges that must be overcome. An appropriate cost and performance balanced strategy will be key to making these materials competitive for use in PV commercial applications.

2.3. Emerging trends in radiative cooling for PV systems

Radiative Cooling (RC) technologies are swiftly advancing, tackling the essential issues of thermal control, efficiency improvement, and longevity in Photovoltaic (PV) systems. New trends in RC emphasize advancements in multi-purpose materials, clear coatings, and the incorporation of cutting-edge PV technologies.

2.3.1. Multi-functional materials

The creation of multi-functional materials is transforming radiative cooling for PV systems, providing notable advancements in thermal regulation and energy efficiency. These materials are designed to enhance mid-infrared emittance for effective heat dissipation while ensuring high solar transmission, resulting in minimal energy losses during use [82]. Composite material stacks, for example, including low-iron soda-lime glass, a porous layer, and diamond coatings achieve a temperature decrease by up to 14 °C under concentrated sunlight into HCPV systems [59]. The layer configurations optimally trade cooling with energy throughput for concentrator sunlight applications. As shown in Fig. 8, the composite material stack consists of multiple layers: a porous low-iron soda lime glass layer (2.28 μm thickness) serves as a broadband anti-reflection (AR) coating to enhance mid-IR emittance; a low-iron soda lime glass layer (500 μm thickness) acts as the primary cooling emitter; and a diamond heat spreader (3 mm thickness) reduces radial temperature gradients while maintaining high solar transmission. Additional layers of low-iron soda lime glass and porous low-iron soda lime glass are included at the bottom to further enhance solar transmission. These layers work together to effectively dissipate heat while

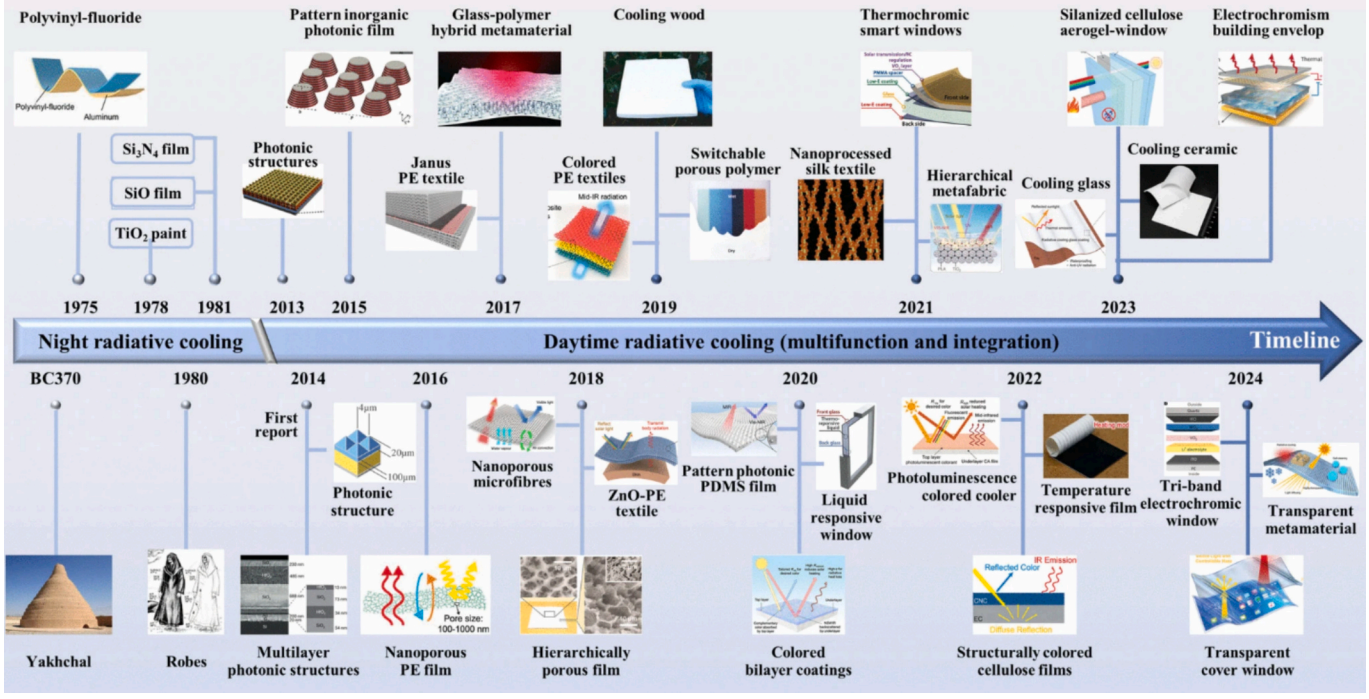


Fig. 7. Timeline of radiative cooling materials, from early passive cooling systems to advanced photonic structures, polymer-based coatings, and smart metamaterials. (Reproduced with permission [196]). Copyright 2024, Springer Nature.

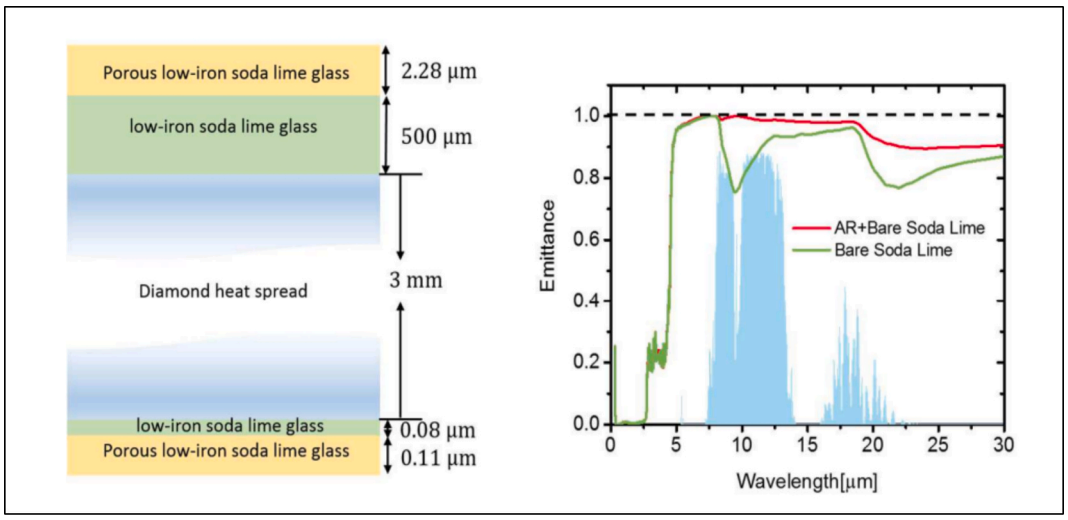


Fig. 8. Cross-sectional schematics of the multi-layer low-iron soda lime glass cooler showing its layered structure and their respective roles in radiative cooling [59].

allowing high solar irradiance to pass through for energy generation.

Similarly, photonic crystal structures, when embedded in materials like low-iron soda-lime glass, control certain light wavelengths to enhance cooling efficiency. These structures can enhance PV diode efficiency by as much as 18 % by selectively reflecting solar radiation and emitting thermal radiation [83]. In addition, RC systems with these advanced materials can even increase low-bandgap photovoltaic cell life by up to 40 % under concentrated solar operation conditions [84]. In addition, nighttime cooling is an essential benefit in that RC systems can even cool the absorber plate temperature by about 9.5 °C below ambient temperature in integrated PV/T-RC systems [85].

In contrast to traditional radiative cooling systems, in which passive heat rejection is achieved without any constraint to energy conversion, PV-integrated cooling systems are required to reduce thermal stress in addition to high solar absorption. The two-function requirement has inspired the creation of new materials and cooling architectures for PV performance improvement with non-compromising energy harvesting [86].

Nanostructured materials and porous radiative coatings enable PV system radiative cooling by optimizing anti-reflectance, light management, and thermal management. Fig. 9 shows that the use of porous PVDF-HFP coatings in PV modules enhances radiative heat dissipation with high solar transmittance. This strategic cooling strategy reduces thermal stress, lowers PV cell temperature, and enhances energy conversion efficiency. The incorporation of a heat pipe also improves passive cooling, such that it emerges as a viable scalable solution for photovoltaic applications. Porous PVDF-HFP coatings, for example, have manifested appreciable cooling impacts in PV applications. A composite coating of Al_2O_3/graphene/PVDF-HFP applied to mini-PV modules achieved a temperature drop of 5.8 °C in outdoor conditions, while the same coating on a PV module provided a cooling effect of 2.4 °C [87]. As indicated in Table 1, research such as Mandal et al. [11] has demonstrated that it is possible to spray-coat or brush-apply porous PVDF-HFP coatings on PV modules for efficient thermal management and minimization of module temperatures by as much as 6 °C. The coatings minimize thermal stress, leading to stable output power and longer PV module lifetimes, especially at elevated temperatures.

The cooling properties of PVDF-HFP coatings are further enhanced

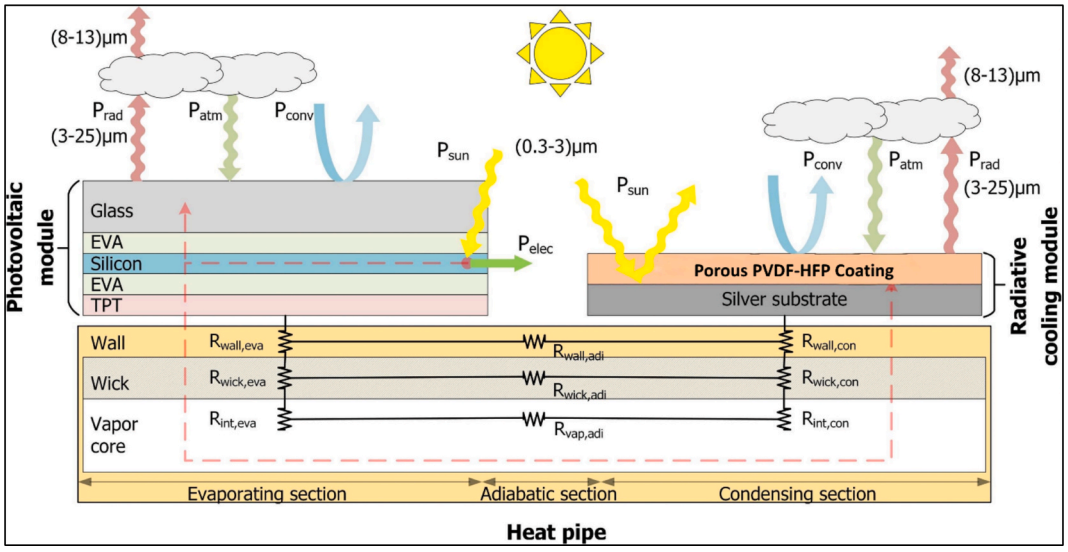


Fig. 9. Schematic illustration of a photovoltaic system coupled with a radiative cooling module. The photovoltaic (PV) module receives solar radiation (0.3–3 µm) for electricity production, and the radiative cooling module, optimized by a Porous PVDF-HFP Coating, emits infrared radiation (3–25 µm) for heat rejection. The heat pipe facilitates heat transfer, optimizing PV module thermal management and improving efficiency (Source: Modified from [197]). Copyright 2017, Elsevier (License Number: 5977980121240).

by their exceptional solar reflectivity and infrared emissivity. The recent literature demonstrates that a PVDF-HFP fiber film, which was fabricated through electrospinning, manifested high solar reflectivity of 99.65 % and emissivity of 90.44 % within the atmospheric window [88]. These properties enable the coatings to reflect the majority of incident sunlight and dissipate absorbed heat efficiently, avoiding excessive temperature rise in PV modules. Durability and stability always constitute key issues for practical use. It is shown through research that PVDF-HFP coatings also exhibit outstanding mechanical strength and long-term stability, and thus they may be applicable for widespread use in industrial PV systems [89]. Additionally, these coatings are of super-hydrophobic nature, which creates a barrier against environmental factors such as rain and dust, also adding to their durability [90].

Porous PVDF-HFP coatings can be synthesized using electrospinning combined with phase separation, which produces a macro-nanoporous structure that increases light scattering and cooling efficiency [15]. Porous P-PVDF films can also be developed through a facile phase separation process, providing low-cost and chemically stable radiative cooling materials [91]. The coating may be directly applied to the surfaces of PV modules through scalable methods like spray-coating or brushing. The coatings have also shown scalability to large commercial and industrial PV systems, with potential for extensive use in energy-efficient photovoltaic systems [87].

Despite these improvements, difficulties remain in optimizing materials for varying atmospheric conditions. Recent research emphasizes enhancing material effectiveness in different climate conditions, such as elevated humidity and reduced solar radiation, since these elements can significantly influence cooling efficiency [201]. Another critical aspect is the seamless integration of RC technology into existing PV systems without much structural modification to increase widespread adoption [85]. Eventually, the economic feasibility of radiative cooling technologies must be confirmed since all the costs involved in their integration must be justified by achievable and measurable benefits in performance, mainly in large-scale commercial applications [84,92]. Overcoming these challenges will be the cornerstone to fulfill the full potential of radiative cooling for photovoltaic systems in the direction of increasing their sustainability and efficiency. The advancement in multi-functional materials forms the basis of efficient RC systems and paves the way to future developments, such as transparent coatings.

2.3.2. Transparent coatings

Multi-functional materials provide the foundation for efficient radiative cooling, and transparent coatings build on this by combining thermal management with high light transmission. These coatings are specifically designed to boost thermal regulation while preserving or improving the Power Conversion Efficiency (PCE) of PV modules.

One of the most important new strategies is transparent radiative heat mirrors made from materials such as indium tin oxide (ITO) and polydimethylsiloxane (PDMS), which transmit visible light while reflecting near-infrared (NIR) photons; it lowers the temperature by 7.2 °C, enhancing the PCE by 0.81 % [93]. Also, hydrogel films based on poly (N-isopropylacrylamide-co-acrylamide) show a high light transmittance over 96 % and an infrared emissivity of 95 %, reducing the temperatures of photovoltaic panels by as much as 9.14 °C [94]. In addition, radiative cooling eco-friendly films constructed from natural silkworm cocoons achieve a temperature difference of 14 °C, enhancing photoelectronic conversion by about 7 % [207].

Anti-reflection radiative cooling (ARRC) glass: integration of anti-reflection and transparent RC layers into a multi-layer structure enhances both the visible light transmittance (0.954) and the mid-infrared emittance (0.951). This design cools multi-crystalline silicon solar cells by 6.6 °C and perovskite cells by 8.6 °C, with an efficiency gain of 1.08 % and 1.79 %, respectively [95]. Moreover, polymeric coatings, including Polyethylene Terephthalate (PET) and PVDF composite films—enhance thermal emissivity and offer other characteristics such as self-cleaning properties and flexibility [204].

Manufacturing complexity, especially for the more advanced coatings such as TRHM and silk-based films, may hinder large-scale adoption [207]. In addition, sustainability issues are driving the development of environmentally friendly materials to reduce environmental impacts. Transparent coatings will also have to be integrated with the existing PV system designs without affecting structural integrity and performance [95,96]. Addressing these challenges will be critical for the broader implementation of transparent coatings in RC technologies. Combining thermal regulation with light transmission, transparent coatings complement material-science advancement and support seamless integration into PV technologies.

2.3.3. Integration with advanced PV technologies

Complementing these material developments, the integration of RC with advanced photovoltaic technologies opens up new perspectives for a higher system performance and durability. The integration of RC and novel PV configurations will help researchers to overcome a number of major challenges related to heat management, efficiency optimization, and system longevity. A notable trend is the integration of RC with photovoltaic-thermal (PV/T) systems, which produce electricity and heat during the day while providing night-time cooling. Studies show that RC integration can reduce solar cell temperatures by up to 9.5 °C below ambient levels and increase exergy efficiency by 0.65 % [97,98,202]. Similarly, hybrid systems, which combine the TE modules with PV systems, take advantage of the cooling potential from TE modules to reduce the temperature by approximately 8 °C, while increasing the cooling performance by 27 % [99].

Advanced materials and full-spectrum designs are also transforming RC integration. Among them, ideally emissive top-layered materials used in flat-plate PV/T collectors have been able to show considerable cooling performance without the need for structural modification. Thermal and electrical efficiency is further improved in full-spectrum radiative cooling strategies by optimizing above-bandgap absorption, sub-bandgap reflection, and thermal radiation. The above methods exhibit significant effectiveness for bifacial photovoltaic (PV) modules, with temperature drops varying from 2.3 to 4.4 °C and power improvements ranging from 7.3 % to 8.4 % [100]. Another new approach is trifunctional systems integrating photovoltaic, photothermal, and radiative cooling technologies (PV-PT-RC). These systems optimize energy conversion and cooling performance, particularly in regions with high solar irradiance. These systems optimize energy conversion and cooling performance, especially in regions with high solar irradiance [97–98]. Moreover, semi-transparent photovoltaic glazing with passive radiative cooling coatings has emerged as an attractive solution for building-integrated PV applications that reduce heat gain by 15 % and increase electricity generation by 3 % [213].

In spite of the advancements, challenges remain on material optimization and design for different climatic conditions, including high humidity or reduced solar irradiation. Ongoing advancement of new materials is essential to guarantee reliable cooling efficiency across different environmental settings [100–102]. Scalability and cost-efficiency are crucial elements, as the hybrid and trifunctional systems must be integrated economically to guarantee their broad usage [97–98,100]. Further investigation of the effects of environmental factors such as wind speed and humidity on RC system performance is also necessary to ensure reliable operation in different geographic locations [100].

Advancements in multi-functional materials, clear coatings, and synergy with cutting-edge PV technologies highlight the transformative promise of RC. Achieving these advantages in extensive solar energy implementations will rely on addressing obstacles related to material efficiency, production scalability, and ecological repercussions. The obstacles ahead for RC technologies involve transforming solar energy and propelling worldwide initiatives towards sustainable and renewable energy solutions.

3. Challenges

Radiative cooling technologies present substantial opportunities to improve the efficiency and effectiveness of Photovoltaic (PV) systems. Nevertheless, various significant obstacles need to be tackled to guarantee their efficient incorporation and expandability. Even with progress in materials and design, challenges like manufacturing scalability, environmental durability, integration obstacles with advanced PV technologies, and issues in economic viability remain. Addressing these challenges is crucial for unlocking the complete potential of radiative cooling in photovoltaic applications. As illustrated in Fig. 10, these challenges can be categorized into four main areas: (1) scalability & fabrication costs, (2) environmental durability & testing, (3) integration constraints, and (4) techno-economic gaps. This part analyzes these challenges, emphasizing problems connected to production methods, inadequate testing in various climate conditions, limitations in combining with new PV technologies, and the absence of comprehensive techno-economic analyses.

3.1. Scalability and fabrication costs

The scalability and integration cost of these RC technologies within PV systems are the main challenges toward their feasibility and adoption. These are driven by the use of special materials, complexity in manufacturing, and the need to balance cost-effectiveness with performance requirements. Radiative cooling technologies depend on materials possessing particular thermal and optical characteristics, including elevated emissivity in the mid-infrared range (8–13 μm) and significant solar reflectivity (0.3–2.5 μm). Advanced materials, including photonic crystals and metamaterials, offer exceptional performance but are constrained by high fabrication complexity and costs, often exceeding $100/m^2 [102–104]. To overcome these limitations, more studies have turned to identify a different form of approximation using low cost and which can be scaled up. It is demonstrated that the recycled glass-based paints and optimum solution-processed polymers such as the PDMS with SiO_2 microspheres help in reduction in the cooling costs and ensured efficient heat discharge. These approaches enable scalable manufacturing at costs as low as $5–10/m^2 [105,106,203].

The economic feasibility of RC technologies is closely tied to fabrication costs and scalability. Transparent coatings, which cost $20–50/m^2, represent a middle ground between high-performance photonic materials and low-cost alternatives such as recycled glass paints. As shown in Table 3, manufacturing techniques vary significantly in terms of cost, scalability, and challenges. Advanced techniques such as photonic crystals demonstrate high performance but are difficult to scale, whereas methods involving recycled glass and PDMS-based materials provide a better balance of cost, scalability, and moderate performance [102,103,105].

Integrating RC technologies with PV systems adds further complexity. Radiative cooling materials must effectively manage thermal loads without hindering PV efficiency. Hybrid systems, such as those combining RC with heat pipes or thermoelectric generators, enhance thermal management but increase fabrication costs due to additional components and integration challenges [93,107]. Despite these challenges, integrating RC technologies offers significant economic benefits, including reduced PV module operating temperatures, enhanced energy conversion efficiency, and extended module lifespans. Studies indicate that RC systems can decrease module temperatures by 10–15 °C, leading to a 1–3 % increase in efficiency and a potential 20–30 % improvement in module lifespan [44,201].

Despite the challenges, RC technologies offer significant long-term economic advantages. By reducing PV module temperatures, RC systems improve energy conversion efficiency and extend the lifespan of PV modules, leading to reduced operational costs. The potential for integration in regions with high solar irradiance and cooling demands further enhances their appeal. Continued innovation in materials and manufacturing techniques is essential to overcome current limitations and facilitate widespread adoption.

3.2. Environmental durability and testing

Radiative Cooling (RC) technologies, particularly when integrated with Photovoltaic (PV) systems, face significant challenges related to environmental durability and the lack of comprehensive testing across diverse climatic conditions. Ensuring these systems perform effectively under various environmental conditions is crucial for their broader adoption and long-term success. One major challenge is the durability of RC materials under harsh environmental conditions. Prolonged exposure to ultraviolet (UV) radiation can degrade the materials, reducing their reflectivity and emissivity over time. Thermal cycling and high ambient temperatures accelerate material aging, leading to a decline in performance. For instance, coatings such as polytetrafluoroethylene (PTFE) and multi-scale particle-doped materials demonstrate effective cooling properties but require enhanced UV resistance to maintain performance over extended periods [108,109]. Additionally, the accumulation of dust and particulates, especially in arid and semi-arid regions, obstructs surface properties and necessitates regular maintenance or the development of self-cleaning surfaces [110].

Humidity presents another significant challenge, particularly in tropical and subtropical regions. High relative humidity interferes with the atmospheric transparency required for effective radiative cooling, as water vapor and aerosols scatter and absorb infrared emissions. This

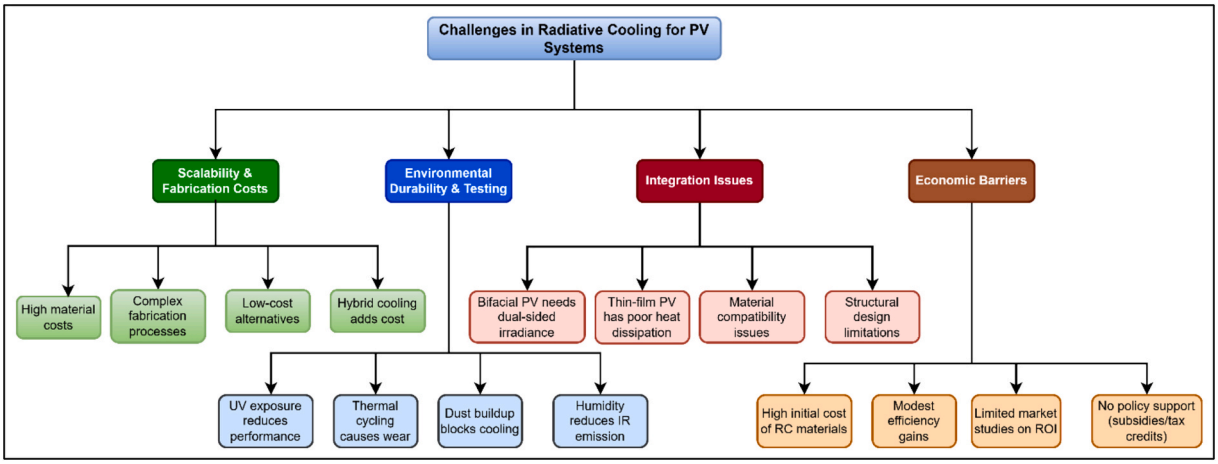


Fig. 10. Challenges in adopting radiative cooling for photovoltaic (PV) systems.

Table 3
Cost comparison of manufacturing techniques.

Manufacturing technique	Cost ($/m^2)	Scalability	Challenges	Performance	References
Photonic crystals	>100	Low	High fabrication complexity	High performance, but limited scalability	[102–104]
Transparent coatings	20–50	High	Material durability issues	Moderate durability under outdoor conditions	[103,105,202]
PDMS with SiO_2 microspheres	5–10	High	Moderate cooling performance	Effective cooling with scalable production	[106,211]
Recycled glass-based paint	5	High	Limited testing in diverse climates	Cost-effective with potential scalability	[105,202]
Hybrid cooling systems	Variable (high)	Low	Integration complexity with PV systems	Enhanced thermal management	[93,107]

effect is particularly pronounced in areas such as Southeast Asia and South America, where performance declines have been observed due to high atmospheric moisture content [110,111]. Despite these limitations, research efforts have shown that using advanced materials such as barium sulfate ($BaSO_4$) coatings can mitigate some of these challenges by optimizing emissivity and reflectivity under diverse environmental conditions [109,112].

The lack of comprehensive testing across a variety of climatic zones further complicates the deployment of RC technologies. Many studies focus on arid regions, where RC performance is inherently favorable due to low humidity and clear skies [110,209]. However, these findings may not translate to other climates, such as humid tropical regions or urban environments with limited sky visibility. For instance, testing in humid regions such as the Mediterranean Basin has revealed reduced performance due to atmospheric scattering effects, underscoring the need for region-specific evaluations [113]. Urban areas also pose unique challenges, as higher ambient temperatures and obstructed sky views diminish cooling efficiency [212]. As shown in Fig. 11, the extent of RC technology testing varies significantly across global regions. While North America, Europe, East Asia, and Australia demonstrate moderate testing activity, regions such as Sub-Saharan Africa, South Asia, and Southeast Asia remain underexplored. In particular, the Middle East and North Africa (MENA) region and Sub-Saharan Africa face a significant lack of testing despite their potential for adopting RC technologies due to their high solar irradiance levels [49,114]. Bridging these gaps requires international collaboration and the development of standardized testing protocols to evaluate the performance of RC technologies in diverse conditions.

Outdoor field tests yield in situ performance data through measurement of climatic impacts like irradiance, humidity, and dust loading. Environmental variation leads to spurious measurements, though, lowering accuracy. Controlled environment laboratory tests with solar simulators and spectroradiometers yield accurate measurements under defined conditions but fail to simulate long-term exposure effects [115]. Standard test techniques in use currently do not comprehensively capture local climatic variability, especially where infrared radiation is impeded by moist or polluted conditions [116]. a countermeasure to such challenges, experimental field testing needs to be performed in less studied areas to validate RC technologies and determine possible areas of optimization. For example, simplified mathematical models with experimentation verification, as evident from test protocols in Brazil, provide affordable means for determining performance in non-standard environments [114]. IEC Technical Specification 63126:2020 sets standards for high-temperature operations without involving time-consuming region-specific test protocols, which creates the need for standards modified based on different climates [117].

Existing research on PV-compatible composites and coatings centers on thermal control, self-cleaning, and weathering resistance. Silica-based anti-reflection coatings provide maximum transmission of light, with hydrophobic coatings having a minimum of dust deposition [118]. Phase-change materials like hexadecanol-palmitic acid/expanded graphite composites ensure optimal thermal stability through maximum lowering of temperatures in PV modules by up to 5.1 °C as well as the increase of efficiency [119]. Additionally, high-emissivity coatings, i.e., hafnium and tantalum carbide coatings, have shown promising performance under harsh climates [120]. Cost-effectiveness and scalability are achieved by advancements in manufacturing, such as roll-to-roll processing, nano-imprinting, and spray-coating. Nano-imprinted polymer matrices improve spectral selectivity, and spray-coated fluorocarbon layers improve surface durability against UV exposure and mechanical

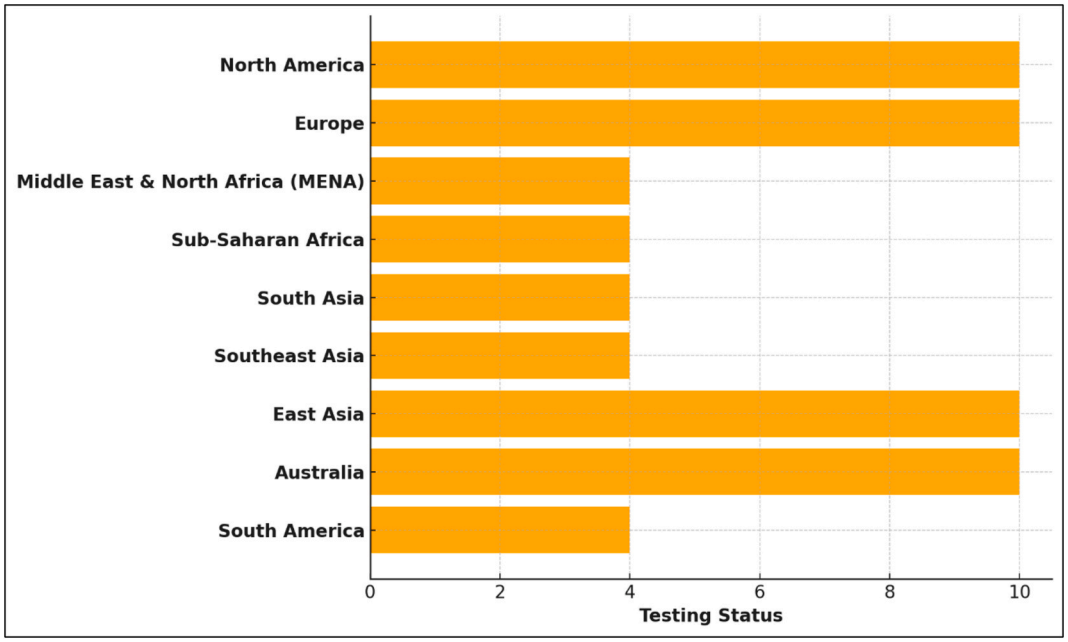


Fig. 11. Bar chart showing regions with significant testing gaps.

stress [121]. These developments collectively enhance economic feasibility and weather resistance, thereby enhancing the applicability of PV-RC integration for large-scale deployment.

PV-RC systems undergo severe climate-induced degradation. In humid climates, the elevated moisture level accelerates corrosion at the expense of durability [122]. In arid regions, dust accumulation reduces irradiance reception, necessitating frequent cleaning [123]. Additionally, UV degradation also increases mechanical wear in polymer-based cooling layers, limiting long-term functionality [124]. Despite these challenges, successful implementations exist. In arid hot climates, nighttime radiative cooling technology has lowered PV module temperatures by 2–7 °C, with a rise in power output [125]. Hybrid cooling systems, which integrate radiative cooling and forced convection, have shown a 13 % improvement in efficiency [126]. Green Roof Integrated PV (GRIPV) systems produce localized cooling, with an improvement in efficiency by 16 % under the peak summer conditions [127].

Addressing the durability and environmental challenges of RC technologies requires a multidisciplinary approach involving materials science, atmospheric modeling, and field testing. Expanding research efforts to regions with significant testing gaps, such as Sub-Saharan Africa and South Asia, will ensure that RC technologies are optimized for diverse climates and environments. By bridging these gaps, RC technologies can be better tailored for regional needs, enabling widespread adoption and integration with PV systems for enhanced energy efficiency and sustainability.

3.3. Integration constraints

The integration of Radiative Cooling (RC) technologies with bifacial and thin-film Photovoltaic (PV) systems presents unique challenges due to the specific requirements of these advanced PV technologies. Addressing these constraints is crucial for optimizing the performance and efficiency of such systems.

One of the primary challenges in combining RC technologies with bifacial PV systems lies in effective thermal management. Bifacial PV systems rely on their ability to capture sunlight from both the front and rear surfaces, which leaves limited surface area for the application of radiative cooling materials. Traditional cooling methods, such as water or air cooling, can obstruct rear irradiance, making radiative cooling a more feasible option for these systems. Studies have shown that full-spectrum radiative cooling strategies can reduce module temperatures by 2.3–4.4 °C, resulting in a power gain of 7.3 %–8.4 %, compared to modest temperature reductions of 1.1–2.2 °C achieved with traditional methods [100,128,129]. Thin-film PV technologies, on the other hand, face distinct challenges due to their compact and lightweight design. While these features are advantageous, they complicate heat dissipation. Integrating RC technologies with thin-film PV systems must not compromise their structural integrity or flexibility, and optimized designs have demonstrated temperature reductions of up to 11.65 °C, significantly enhancing cooling power and electrical efficiency [27,130].

Material compatibility is another critical factor influencing the integration of RC technologies with PV systems. Radiative cooling materials must exhibit high emissivity in the mid-infrared spectrum (8–13 μm) to effectively dissipate heat while maintaining high solar transmission. This ensures that the cooling effect does not interfere with the light absorption necessary for energy conversion. Advanced coatings, such as polycarbonate films, have proven effective for bifacial PV systems, as they maintain cooling performance without obstructing rear irradiance. For thin-film PV systems, the use of multilayer structures like Polyethylene Glycol Terephthalate (PET) films helps balance reflectivity and cooling efficiency, but these designs often introduce additional complexity and cost [27,129,130].

While radiative cooling can improve the overall efficiency of PV systems, the degree of efficiency gains varies between bifacial and thin-film technologies. In bifacial PV systems, the integration of radiative cooling must carefully balance the dual-sided light capture with cooling performance. For thin-film PVs, the higher temperature sensitivity of the materials makes them more responsive to radiative cooling, enabling greater efficiency improvements. However, achieving these gains requires careful alignment of cooling materials with the unique spectral properties of each technology. Research indicates that while the cooling effect in thin-film PV systems can be substantial, the design and integration must address inherent material limitations [100,130,131].

Structural and design constraints also play a significant role in implementing RC technologies. In bifacial PV systems, the placement and orientation of cooling materials are crucial to avoid blocking reflected sunlight, which is vital for dual-sided operation. Dual-surface cooling concepts, which utilize radiative cooling on both sides of bifacial modules, require innovative geometric designs to optimize thermal and electrical performance. Thin-film PV systems, by contrast, prioritize lightweight and adaptable designs. The integration of cooling technologies must align with these priorities, ensuring that they do not add significant weight or compromise flexibility [46,131,132].

Finally, environmental and economic factors must be considered. Water-based cooling methods, although effective, are less suitable for arid regions due to their high-water consumption. In contrast, radiative cooling operates without additional energy or water usage, making it a more sustainable solution. The cost of advanced cooling materials, such as high-emissivity coatings and multilayer films, remains a challenge. However, simpler and more stable designs for radiative cooling can help reduce costs and improve scalability, enabling broader adoption across different PV technologies. In summary, the integration of radiative cooling technologies with bifacial and thin-film PV systems requires addressing challenges related to thermal management, material compatibility, structural design, and economic feasibility. By tackling these constraints, radiative cooling technologies can be effectively implemented to enhance the overall performance and efficiency of advanced PV systems.

3.4. Techno-economic gaps

Radiative Cooling (RC) technologies integrated with Photovoltaic (PV) systems offer substantial potential to improve system performance by reducing operating temperatures and enhancing energy efficiency. However, significant techno-economic gaps hinder their widespread adoption and scalability. These challenges revolve around high initial costs, economic viability, and the absence of comprehensive market studies, which are essential for assessing the long-term feasibility and benefits of these integrated systems.

One of the primary economic challenges is the high initial cost of integrating RC technologies with PV systems. Advanced materials with high emissivity and reflectivity, required for effective radiative cooling, often involve complex manufacturing processes and material costs. For example, multilayered coatings such as Polyethylene Glycol Terephthalate (PET) films and polycarbonate films are effective but expensive, limiting their adoption in large-scale PV installations [102,129]. Additionally, incorporating RC technologies into PV systems demands engineering expertise to maintain the balance between thermal management and energy conversion, further increasing the initial costs [46,101].

The economic viability of RC technologies also depends on their ability to deliver significant performance improvements under real-world conditions. Although RC systems can reduce operating temperatures and extend the lifespan of PV modules, the efficiency gains are often modest. For instance, studies have shown that enhanced RC in a photovoltaic-thermal (PVT) system reduced module temperatures by only 2 °C, resulting in a total exergy efficiency improvement of 0.65 % [201]. Similarly, the Levelized Cost of Energy (LCOE) for integrated RC-PV systems can be prohibitively high, as seen in solar thermal cooling systems with an LCOE of $2.35/kWh compared to $0.9/kWh for conventional PV systems [133]. These cost-benefit limitations highlight the

need for optimizing RC designs to achieve more significant economic and operational advantages.

Another critical gap is the absence of comprehensive market studies evaluating the long-term economic and environmental benefits of RC technologies in PV systems. Currently, most research focuses on technical aspects, such as material development and cooling performance, while market-oriented studies remain scarce. This lack of market analysis hampers the ability of stakeholders to assess the scalability and profitability of RC technologies. For example, detailed life cycle cost analyses and environmental impact evaluations are essential to quantify the overall benefits of RC-PV integration, including reduced carbon emissions and operational savings [134]. Without such studies, policymakers and investors may lack the confidence to support or fund these technologies, slowing their adoption [135]. Furthermore, the absence of market studies also impacts technological development and innovation. Market studies often identify opportunities for improvement, guiding researchers toward optimizing designs and addressing performance bottlenecks. For instance, comprehensive data on RC performance in different climatic conditions and its impact on PV efficiency could lead to more targeted research and development efforts [136]. Additionally, market studies can facilitate the development of industry standards and regulations, which are critical for establishing a competitive and sustainable market for RC technologies [137,138].

The lack of policy support and incentives is another significant consequence of the techno-economic gaps. Policymakers rely on market studies to understand the potential impact of emerging technologies and develop supportive policies. Without this information, it is challenging to secure subsidies or tax incentives that could drive the adoption of RC technologies. For example, successful policy frameworks for solar PV systems, such as feed-in tariffs and renewable energy credits, have been instrumental in driving their growth. Similar mechanisms could be extended to RC-PV systems to address economic barriers [133,139]. However, while RC technologies integrated with PV systems hold promise for improving thermal management and energy efficiency, the techno-economic gaps—such as high costs, limited performance improvements, and the lack of comprehensive market studies—pose significant challenges to their adoption. Addressing these gaps through detailed techno-economic assessments, cost-effective material development, and supportive policies is essential to realize the full potential of RC technologies in the renewable energy landscape.

4. Opportunities

The intersection of material science, manufacturing innovation, and system integration presents a wealth of opportunities to address current challenges in radiative cooling and Photovoltaic (PV) technologies. As advancements in materials and manufacturing methods continue to evolve, the potential for scalable, cost-effective solutions grows significantly. Additionally, comprehensive testing frameworks and techno-economic models are essential to validate performance and ensure sustainability. As illustrated in Fig. 12, these opportunities can be categorized into five main areas: (1) advances in material science, (2) scalable manufacturing processes, (3) comprehensive testing frameworks, (4) techno-economic models, and (5) future integration with next-gen PV. This section explores the avenues for innovation, focusing on materials, processes, testing, and integration with next-generation PV systems.

4.1. Advances in material science

Advancements in material science have opened up significant opportunities to enhance the efficiency and scalability of radiative cooling and PV technologies. These innovations focus on designing materials with optimized properties to address critical performance barriers while supporting large-scale adoption. One major opportunity lies in high solar reflectance materials, which can reflect up to 90 % of solar radiation to minimize heat absorption, making them ideal for passive cooling systems. This property reduces the thermal stress on PV modules, enhancing their efficiency and lifespan, especially in regions with high solar irradiance [140,141]. Similarly, high infrared emissivity, essential for radiative cooling, enables effective heat dissipation within the atmospheric window (8–13 μm). Materials with emissivity levels of 93–96 % offer substantial cooling benefits, even under intense sunlight [142,143]. The opportunity to develop durable and weather-resistant materials has also advanced significantly. Hybrid polymer-based coatings and superhydrophobic materials with self-cleaning capabilities ensure long-term performance by resisting environmental degradation. These features are crucial for outdoor applications, especially for PV systems and building envelopes, where exposure to harsh conditions is inevitable [144,145].

Cost-effective and scalable solutions are another key area of opportunity. Materials such as photonic structures and bioinspired designs,

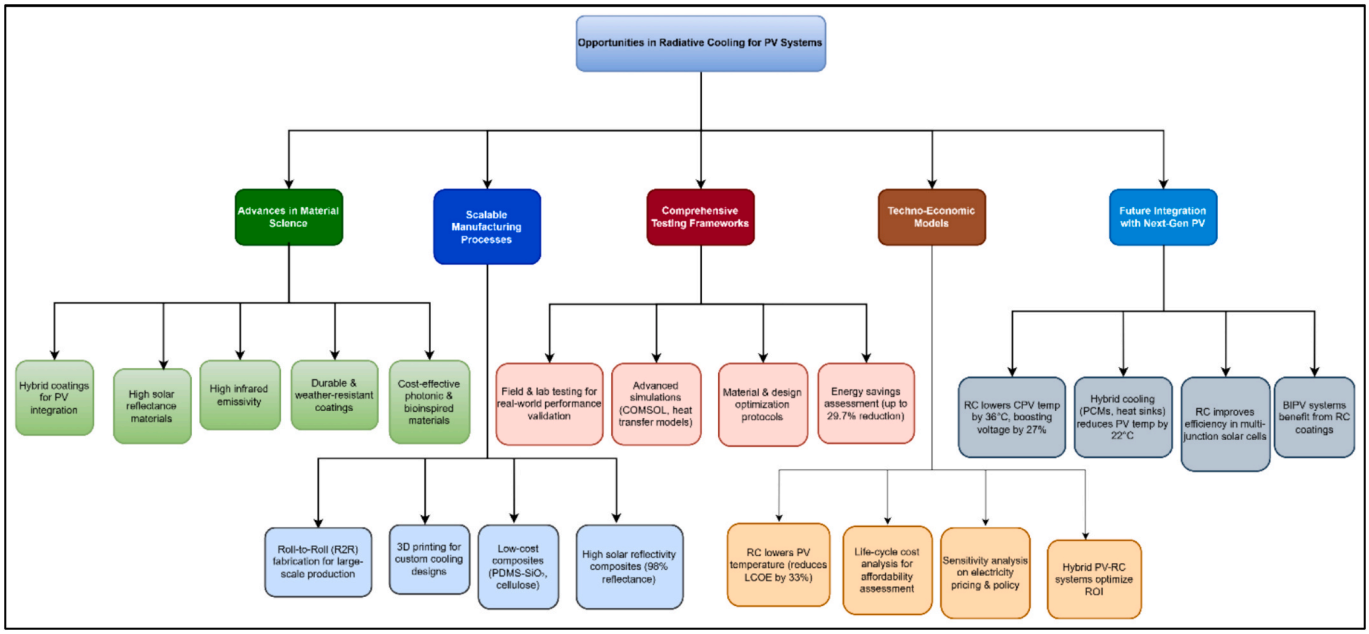


Fig. 12. Opportunities in radiative cooling for photovoltaic (PV) systems.

which leverage advanced fabrication techniques, offer innovative ways to improve solar reflectance and infrared emission. These materials provide the potential for large-scale deployment in both urban and industrial environments [146,143]. Similarly, cementitious materials adapted with radiative cooling properties offer low-cost alternatives for integrating cooling systems into building construction, addressing both affordability and functionality [147]. Finally, the ability to integrate radiative cooling with PV systems presents transformative opportunities for energy efficiency. By combining cooling materials with PV modules, systems can achieve enhanced thermal management, leading to improved energy conversion rates and extended operational lifetimes. Such integration is especially critical as PV technologies evolve to meet increasing energy demands sustainably.

4.2. Scalable manufacturing processes

Scalable manufacturing processes are essential for realizing the widespread adoption of radiative cooling and Photovoltaic (PV) technologies. By leveraging techniques such as roll-to-roll (R2R) manufacturing, 3D printing, and low-cost composites, these technologies can be produced efficiently and cost-effectively, driving innovation and sustainability. Roll-to-roll (R2R) manufacturing is a highly scalable and cost-efficient process that enables continuous production of thin films on flexible substrates. This method is ideal for producing radiative cooling materials and PV films in large areas, suitable for integration into building facades or solar modules. R2R processes significantly reduce material waste and energy consumption during production, making them environmentally friendly and economically viable. For instance, R2R manufacturing can create thin radiative cooling films that enhance PV module efficiency by reducing operating temperatures, ultimately extending the lifespan of these systems [106]. Additionally, this technique allows for high production throughput, ensuring rapid deployment of advanced energy solutions.

3D printing offers unique opportunities for the customization and optimization of radiative cooling and PV technologies. This technique allows for the precise fabrication of complex geometries that maximize the performance of cooling materials and PV modules. By enabling rapid prototyping, 3D printing supports localized and on-demand production, reducing logistics costs and material waste. For instance, intricate cooling structures designed using 3D printing can enhance emissivity and reflectivity, contributing to efficient thermal management. Furthermore, conformal printing techniques enable the seamless integration of cooling features directly into PV systems, optimizing both efficiency and durability [148,216]. The ability to produce lightweight, tailored components makes 3D printing a transformative tool in the renewable energy industry.

The use of low-cost composites further advances scalable manufacturing by reducing production costs without compromising performance. Materials such as polydimethylsiloxane (PDMS) with SiO_2 nanoparticles and cellulose-based composites are affordable and easy to process using techniques like extrusion and molding. These composites exhibit high solar reflectance and infrared emissivity, essential for effective radiative cooling. For instance, cellulose-based composites can achieve solar reflectivity of up to 98.44 % and infrared emissivity of 98.7 %, resulting in significant cooling performance under direct sunlight [84,149]. Additionally, these materials are suitable for mass production and can be applied to various surfaces, making them ideal for building-integrated applications and large-scale energy systems. Scalable manufacturing processes, combining R2R production, 3D printing, and low-cost composites, present a transformative opportunity to advance radiative cooling and PV technologies. By reducing costs, improving efficiency, and supporting large-scale deployment, these methods are driving the next generation of sustainable energy solutions.

4.3. Comprehensive testing frameworks

Comprehensive testing frameworks are vital for unlocking the full potential of integrated Radiative Cooling (RC) and Photovoltaic (PV) technologies. These frameworks ensure that combined systems meet performance, reliability, and sustainability standards under diverse environmental conditions, enabling their broader adoption in energy-efficient applications. Testing frameworks offer opportunities to validate the performance of PV-RC systems by focusing on key metrics such as temperature reduction, cooling power, and efficiency gains. Radiative cooling has been shown to lower PV operating temperatures by up to 10 °C, resulting in efficiency improvements of 5–7 % in concentrated PV systems. Such reductions in thermal stress not only enhance energy conversion efficiency but also extend the lifespan of PV modules, addressing a critical challenge in solar energy systems [59,84].

Performance validation requires robust experimental setups that simulate real-world conditions. Outdoor and controlled laboratory testing are crucial for evaluating integrated systems under varying ambient temperatures, solar irradiance, and wind speeds. Comparative testing of PV modules with and without RC integration helps quantify the benefits of thermal management, such as reduced degradation rates and improved durability [201]. These evaluations enable researchers and manufacturers to fine-tune system designs for maximum performance across different climatic zones. Advanced modeling and simulation tools further enhance the development of PV-RC systems by predicting their behavior under complex heat transfer scenarios. Thermal models simulate radiative, convective, and conductive cooling processes, while software such as COMSOL Multiphysics allows for detailed finite element analysis. These tools optimize material properties and system designs, ensuring that RC technologies achieve high emissivity and reflectivity without compromising PV module performance [211].

Material and design optimization is another critical focus of comprehensive testing frameworks. High-emissivity materials, such as photonic crystals or porous layers, play a pivotal role in enhancing cooling performance. Testing protocols must evaluate the optical and thermal properties of these materials to ensure effective cooling in diverse applications. Innovative structural designs further maximize the benefits of radiative cooling by improving heat dissipation and system integration [150]. Additionally, testing frameworks assess the energy savings and environmental benefits of PV-RC systems. By reducing reliance on active cooling methods, these systems can achieve energy savings of up to 29.7 %, significantly lowering operational costs and carbon footprints [206]. These environmental assessments highlight the role of integrated PV-RC technologies in advancing global sustainability goals. However, comprehensive testing frameworks are instrumental in realizing the opportunities offered by integrated radiative cooling and photovoltaic technologies. By combining experimental validation, advanced modeling, and material optimization, these frameworks ensure that PV-RC systems deliver enhanced efficiency, extended durability, and significant energy savings, paving the way for their widespread adoption in sustainable energy applications.

4.4. Techno-economic models

Techno-economic models offer a robust framework to evaluate the feasibility of integrating Radiative Cooling (RC) and Photovoltaic (PV) technologies, focusing on their economic and technical benefits. RC enhances PV performance by reducing operating temperatures, leading to a significant improvement in open-circuit voltage and extending module lifespans. This integration reduces the Levelized Cost of Electricity (LCOE) by as much as 33 % in concentrated photovoltaic systems, showcasing its cost-effectiveness. Additionally, hybrid systems combining PV with thermoelectric generators (TEGs) or solar thermal collectors further optimize energy efficiency and minimize operational costs. Life cycle cost analyses reveal that RC systems, with an energy cost

of $0.011 per kilowatt-hour, are more economical than PV systems in certain applications, which average $0.034 per kilowatt-hour. These findings highlight RC's suitability for regions with high cooling demands, such as arid climates, while PV technologies excel in areas with abundant solar irradiance [84,151,152].

Long-term cost projections in techno-economic models incorporate critical parameters like initial capital costs, operational expenses, system efficiency, and energy yield. Models account for future reductions in costs through learning rates and experience curves, predicting substantial cost savings over time. For instance, investment and dispatch models simulate economic value changes at high PV penetration, while sensitivity analyses assess impacts of varying economic factors like electricity prices and policy incentives. Such models also consider environmental benefits, such as CO_2 emission reductions, enhancing the overall sustainability profile of RC and PV systems. Furthermore, material advancements, such as polycarbonate films with high emissivity, coupled with policy support, have the potential to accelerate the adoption of these integrated technologies globally. As demonstrated in comparative studies across climatic regions, hybrid PV-RC systems are not only environmentally friendly but also economically competitive, offering an attractive pathway for achieving energy sustainability [151,153,154].

4.5. Future integration with next-generation PV systems

Radiative Cooling (RC) presents a transformative opportunity for enhancing the efficiency and longevity of next-generation Photovoltaic (PV) systems through passive and sustainable thermal management. By dissipating heat via infrared radiation, RC effectively reduces the operating temperatures of PV modules, thereby mitigating temperature-induced efficiency losses. For example, RC has been shown to lower PV temperatures by up to 36 °C in concentrated photovoltaic (CPV) systems, leading to a 27 % relative increase in open-circuit voltage and extended system lifespan by 4 to 15 times [155,156]. In conventional PV modules, RC reduces temperatures by approximately 7.2 °C, resulting in a 0.81 % increase in power conversion efficiency [93]. These temperature reductions not only improve energy output but also significantly decrease thermal stress, ensuring longer operational lifespans. Integrating RC with hybrid cooling strategies, such as heat sinks or phase change materials (PCMs), has demonstrated even greater benefits, including a maximum temperature reduction of 22 °C and an 8.57 % increase in power output [157].

RC's adaptability makes it particularly suited for advanced PV technologies, including multi-junction solar cells and building-integrated PV systems (BIPV). For multi-junction cells, combining RC with other cooling techniques reduces the required cooling power by 90 % and achieves a 2 % absolute efficiency gain [155]. Meanwhile, applying RC coatings on BIPV modules mitigates temperature-related efficiency losses while delivering additional cooling benefits. Future innovations in RC materials, such as high-emittance coatings and photonic structures, promise further performance gains. Systems like the NSS-RC-PV/T, which dynamically switch between PV and RC modes, have demonstrated a 33.4 % increase in annual energy output, underscoring the potential for flexible energy management in diverse applications [210]. As the integration of RC into PV systems evolves, its passive nature, minimal energy requirements, and compatibility with other renewable cooling methods ensure its vital role in advancing next-generation PV technologies.

5. Future directions

The integration of radiative cooling and Photovoltaic (PV) technologies is vast in increasing the efficiency and performance of the energy and systems, especially in sustainable energy technologies. Although tremendous progress has already been made so far, there is still much that can be optimized in maximizing the design, performance, and scalability of such systems. The principal research and development avenues for subsequent studies are discussed in this section, including developing strategic roadmaps, multi-functional cool materials through innovation, maximum dual-surface cooling mechanisms, techno-economic model building capable of withstanding the harshness of analysis, and promoting industry alignment towards field deployment. Solutions to these areas of research will best position radiative cooling and PV technology to meet international energy and climate actions targets.

5.1. Roadmap and prioritization of research efforts

Large-scale radiative cooling photovoltaic (RC-PV) integrated system development requires a carefully designed research approach through material innovations, hybrid cooling configurations, techno-economic evaluation, and multi-disciplinary integration. Despite promising advancements, some key gaps in the necessary research areas remain to be filled. These bottlenecks will be overcome to realize the commercial feasibility of RC-PV systems with improved energy efficiency, economic competitiveness, and environmentally sustainable long-term concerns.

One of the primary research gaps is maximizing the performance of RC-PV systems in humid environments. The radiative cooling effect can be minimized by high humidity due to the limitation of the atmospheric transparency window, affecting heat dissipation. To reverse this, there needs to be research on nanostructured coatings with a basis on materials such as phase change materials (PCMs) and nanoparticles to ensure high thermal conductivity and stability. For instance, a three-layer nano-PCM arrangement was shown to reduce PV module temperatures by up to 5 °C and enhance electrical efficiency by 4 % [158]. Furthermore, hybrid nanofluids used in cooling ducts have also shown enhanced convective heat transfer and are thus appropriate to use in humid climates where natural radiative cooling is not as effective [159].

Another pivotal area of research deficiency is the identification of better hybrid cooling designs that optimize cost, life, and efficiency in various climate zones. Hybrid cooling systems that integrate RC with water cooling, thermoelectric cooling, or phase-change materials have been found to be variable in performance. In hot and humid climates, a combination of water and air cooling has demonstrated a rise of 4.5 % in PV efficiency [160]. In arid regions, jet impingement cooling and spectrum filtering have reduced PV temperatures significantly without efficiency loss [32]. These studies highlight the necessity of creating climate-specific hybrid cooling systems that optimize performance under different environmental conditions.

Techno-economic feasibility of using RC in large-scale solar farms compared to rooftop solar systems is another very important research topic. Large-scale solar farms benefit from economies of scale that minimize the cost per unit of electricity generated. They are extremely costly and involve tremendous amounts of infrastructure capital and regulatory approval [161]. Conversely, PV systems atop buildings enjoy front-end energy savings due to smaller cooling loads and are therefore favored for residential as well as commercial buildings. Studies show that RC is cheaper than traditional cooling systems in rooftop applications due to its passive nature and low maintenance [151]. Future research needs to explore the mechanisms through which policy incentives, financing structures, and regulatory support can encourage RC-PV uptake at both scales.

To systematically work towards filling such research gaps, a 3–5-year phased roadmap is suggested that will guide work in the areas of optimization, development, and standardization. Work during the short term (1–2 years) will focus on optimizing Radiative Cooling (RC) materials through the development of natural fiber composites (NFCs) that are economically viable, sustainable, and lightweight [162]. Medium-term (3–4 years) will focus on hybrid RC-PV cooling systems integrating thermal storage, phase change materials (PCMs), and predictive AI-driven controls to raise efficiency and extend longevity [163]. In the long term (5 + years), the goal is to create standardized frameworks for

RC-PV integration, enabling global scalability and regulatory compliance through collaboration with IEC, IEEE, and ISO, capitalizing on solar PV standardization efforts [164]. Fig. 13 illustrates this proposed phased roadmap, outlining the key objectives for each phase, from material innovation to mass adoption.

Interdisciplinary collaboration is crucial for advancing RC-PV technology. AI-based predictive models such as Support Vector Machines (SVM), Gradient Boosting, and Hybrid MLP-CNN have been effectively used to optimize radiative cooling materials by predicting emissivity, thermal efficiency, and degradation under different conditions [165,214]. Machine learning has also accelerated thermal metamaterials, phase change materials (PCMs), and nanomaterials, which speed up material discovery and cooling performance [166]. Additionally, AI-driven RC technologies are also being incorporated into national energy codes and building policy, enabling scalability and policy compliance for sustainable cooling solutions [167,168]. By leveraging AI and interdisciplinary collaboration, RC-PV systems can achieve greater efficiency, durability, and widespread adoption.

In conclusion, the successful deployment of RC-PV systems requires a systematic and multidisciplinary research approach. Nanostructured coatings, hybrid cooling technologies, and techno-economic analysis will drive innovation in the field. Strategic collaborations between academia, industry, and policymakers, however, will ensure that RC-PV systems are strongly integrated into the global energy transition. With a clear roadmap and strong research priorities, radiative cooling can evolve from an emerging technology to a mainstream solution for sustainable energy generation.

5.2. *Multi-functional cooling materials*

The development of multi-functional cooling materials represents a transformative step toward integrating Radiative Cooling (RC) with Photovoltaic (PV) systems to enhance efficiency, durability, and sustainability. Future research should prioritize the creation of scalable, cost-effective materials that not only cool PV systems but also offer self-cleaning, anti-contamination, and adaptive thermal regulation. These advancements will play a critical role in improving long-term performance and reducing maintenance costs in solar energy applications.

One promising avenue is the development of coatings with switchable cooling and heating modes, which would enable year-round efficiency across diverse climates. For instance, dynamic phase-change materials (PCMs) have shown potential for adaptive thermal regulation, allowing PV modules to cool in summer while retaining heat in winter [169]. Advances in bio-inspired radiative cooling materials, such as cellulose-based coatings and IR-adaptive materials, have also demonstrated the ability to maximize thermal radiation efficiency while maintaining structural durability [170].

The use of nanomaterials has emerged as a key solution to improve thermal management and self-cleaning capabilities in PV panels. Hydrophobic and photocatalytic coatings utilizing TiO_2 and SiO_2 nanoparticles can minimize dust accumulation, ensuring that panels maintain high efficiency over extended periods [171]. These coatings also enhance the anti-reflective and self-cleaning properties of solar panels, reducing the need for frequent maintenance in harsh environments [172].

Additionally, the integration of MXenes and aerogels into radiative cooling materials presents an opportunity to develop lightweight and durable passive cooling solutions. These materials exhibit high surface area and thermal conductivity, making them ideal for hybrid PV-thermal cooling systems [173,174]. Spectrally selective materials, including metamaterials, are also gaining traction, as they can enhance cooling efficiency by selectively reflecting solar radiation while emitting thermal energy through the atmospheric transparency window [175].

For successful commercialization, scalability and cost-effectiveness remain major challenges. Researchers are exploring high-throughput, roll-to-roll manufacturing techniques to lower production costs and improve material uniformity, similar to advances in thin-film solar technologies [13]. Additionally, standardizing material performance

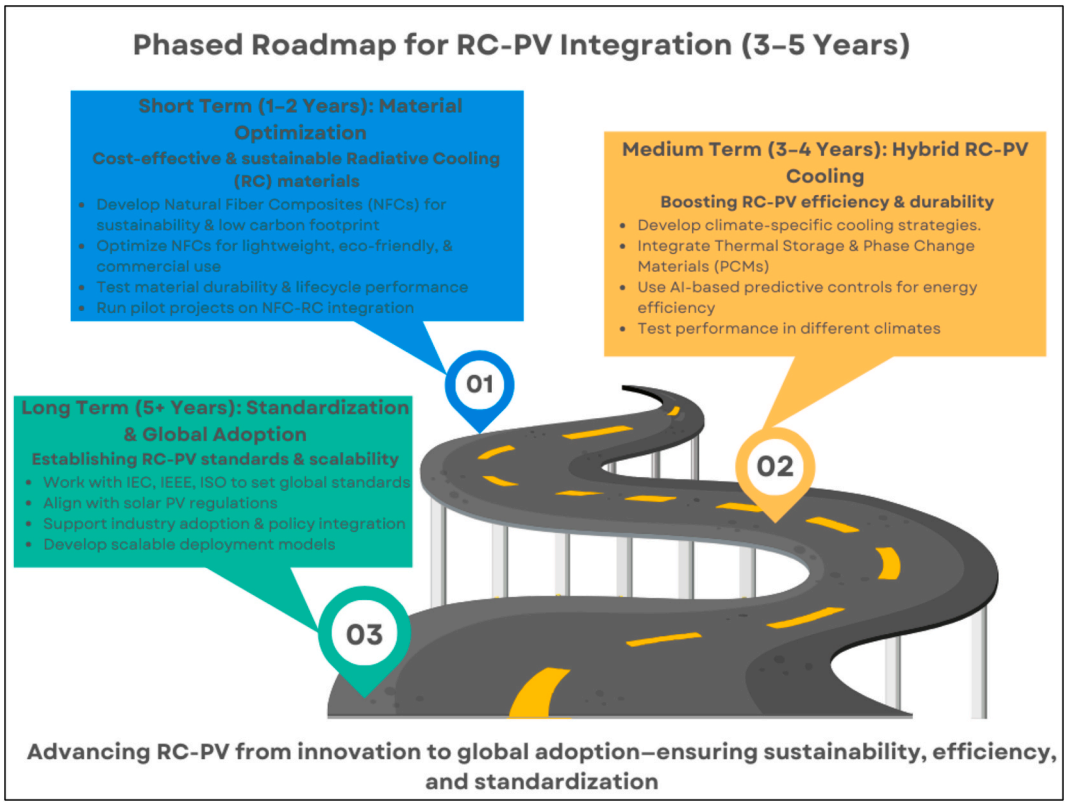


Fig. 13. Proposed 3–5-year roadmap for RC-PV optimization, cooling, and standardization.

metrics through international guidelines, such as ISO and IEC standards, will be essential for market adoption and regulatory compliance [102].

As these advancements materialize, multi-functional cooling materials will not only enhance energy efficiency in solar PV but also pave the way for their integration into sustainable architecture, urban cooling systems, and next-generation renewable energy technologies. Future research should focus on improving material longevity, real-world validation through pilot studies, and optimizing hybrid cooling configurations for different climate conditions.

5.3. Dual-surface cooling strategies

Dual-surface cooling strategies leverage radiative cooling in conjunction with additional heat dissipation mechanisms to optimize thermal management in Photovoltaic (PV) systems. The hybrid cooling methods aim to minimize temperature variations, increase PV efficiency, and prolong system longevity by using front and back surface cooling approaches. With the utilization of bifacial PV modules, Thermophotovoltaic (TPV) systems, and hybrid cooling methods, the dual-surface cooling techniques provide scalable solutions for various climatic environments.

One of the most significant ones is bifacial PV module radiative cooling, where both sides are provided with heat dissipation to achieve optimal performance. Research has proven that self-cooling coated bifacial modules can operate at lower temperatures, and thus they receive 7.3 %–8.4 % more power output per year [100]. Additionally, phase change materials (PCMs) such as calcium chloride hexahydrate have been tested as a cooling medium for bifacial PV, showing up to 10.7 % power gains in certain conditions [176]. This indicates that passive radiative cooling can be combined with new materials to further improve bifacial PV cooling methods.

Another innovative approach involves dual-layer thermal coatings designed for emissivity-tunable cooling. These coatings selectively reflect solar radiation while maximizing thermal dissipation, reducing the steady-state temperature of PV modules by up to 6.02 °C [129]. Spectrally selective materials, including metamaterial thermal emitters, have also been integrated into hybrid PV-Thermal (PV/T) systems, where heat absorption and dissipation are precisely controlled to enhance energy conversion efficiency [177].

For Thermophotovoltaic (TPV) applications, dual-surface cooling plays a critical role in optimizing energy conversion. Plasmonic metamaterial selective emitters can be designed to enhance near-infrared absorption while suppressing unnecessary thermal losses, ensuring high-efficiency operation in TPV systems [178]. Meanwhile, hydrogel-based rear-surface cooling solutions have demonstrated potential for further improving PV performance through moisture harvesting and evaporative heat dissipation [179].

The feasibility of dual-surface cooling strategies extends beyond technical advancements to include economic viability and scalability. Hybrid PV-Thermal-Radiative Cooling (PV/T-RC) systems, which switch between cooling and heating modes depending on seasonal requirements, have been shown to optimize year-round energy efficiency [163]. However, scaling up such technologies to massive PV plants involves further research in cost-effective materials, best manufacturing practices, and incorporation into current PV infrastructure [151].

Future studies should focus on on the development of adaptive cooling materials that operate dynamically according to environmental conditions, including emissivity-switchable coatings and phase-changing hybrid materials. Furthermore, scalability of innovative coatings must be enhanced with cost-effectiveness and durability for business implementation. New hybrid configurations combining radiative cooling, evaporative cooling, and spectral-selective emission are also need to be explored to further enhance multi-functional energy management optimization. Finally, techno-economic compromises need to be addressed by integrating low-cost but high-quality technology into industrial production for large-scale deployment of PV. By systematically addressing these research gaps, dual-surface cooling strategies can be commercialized effectively, leading to higher efficiency, lower maintenance costs, and more sustainable solar energy solutions.

5.4. Techno-economic models

The integration of Radiative Cooling (RC) with Photovoltaic (PV) systems is one potential way to improve energy efficiency, minimize the cost of operation, and minimize the Levelized Cost Of Electricity (LCOE). However, the widespread use of RC technologies depends on structured economic models that evaluate feasibility, financial risk, and policy incentives. Some established frameworks, including Multi-Criteria Decision Making (MCDM), Cost-Benefit Analysis (CBA), and LCOE modeling, help in making investment in RC-PV systems. MCDM is widely used in renewable energy planning to allow decision-makers to evaluate competing technologies against multiple criteria, such as cost-effectiveness, efficiency, and environmental factors. The Analytic Hierarchy Process (AHP) of widely used MCDM techniques has been applied to renewable energy projects for the support of financial decision and technology selection [180]. Additionally, CBA models account for the economic viability of RC-PV integration on the basis of balancing the initial investment and long-term operation costs. Studies show that temperature of the PV module can be lowered by as much as 36 °C, which amounts to 27 % open-circuit voltage augmentation and a 4 to 15-times prolongation of system life [181].

Though savings on individual LCOE for RC-PV systems are limited in existing literature, passive cooling technologies conserve $0.01–$0.03 per kWh and RC is thus an economically viable option in high solar irradiance zones [182]. Payback periods for companies are from 5.32 to 7.89 years depending upon climatic conditions, cost of electricity, and available incentives [49]. Investors must balance long-term cost advantages, such as reduced maintenance, increased module efficiency, and longer system lifespan, all of which enhance the return on investment (ROI). To facilitate widespread adoption, various financing models can help mitigate upfront investment challenges. One such structure is the Power Purchase Agreement (PPA), in which companies pay only for produced electricity and not for the PV system, thereby limiting capital expenditure [183]. Similarly, the Pay-As-You-Go (PAYG) system allows businesses to pay as they go over time based on the energy used, thus making RC-PV systems financially viable for small and medium-sized businesses [184]. In addition, Property Assessed Payments for Energy Retrofits (PAPER) programs utilize local improvement charges to finance RC-PV projects so that energy savings every year will exceed the loan repayment cost, hence the investment being cost-neutral in essence [185]. Governments can also encourage adoption through Green Bonds and Venture Capital (VC) investments, already successfully used in Europe and China to finance renewable schemes [208].

Beyond financing, public–private partnerships and policy incentives are required for extensive RC-PV deployment. Adoption can be enabled by policymakers imposing tax credits, direct subsidies, and regulatory obligations, similar to those for growing solar PV. For instance, California's energy efficiency standards for Title 24 require cool roof materials, and similar policies could be enacted for enabling RC integration [186]. They must also invest in R&D collaborations to achieve optimized RC material efficiency, long-term performance durability, and compatibility with hybrid PV systems [187].

A climate-based deployment strategy is also necessary to optimize RC-PV efficiency and cost-effectiveness. Tropical regions, such as Singapore, are benefited by sub-ambient cooling advantages of up to 2 °C, with RC being a feasible energy-saving solution [188]. Humid subtropical areas like Hong Kong and Southern China have effectively employed RC coatings that achieve temperature drops of up to 2.7 °C, even in moist conditions [189]. Meanwhile, arid and semi-arid regions such as the Middle East and U.S. Southwestern deserts have demonstrated up to 13 °C PV module cooling through hybrid PV-heat pipe-

thermoelectric generator systems, with a 7.2 % improvement in energy yield [190]. Additionally, Low-latitude coastal regions in China have also demonstrated temperature decreases of 6.5 °C under 750 W/m^2 solar radiation, even in humid environments [189]. In Pakistan, RC coatings lowered annual cooling loads by 76.01–112.77 kWh/m^2, equating to 30.40–45.11 kWh/m^2 electricity savings [49]. The findings indicate regional variability in RC performance, further advocating for the use of climate-adaptive deployment strategies.

In conclusion, large-scale deployment of RC-PV systems depends on economic modeling, funding mechanisms, and policy incentives promoting universal adoption. High-temperature, high-sun regions ought to be prioritized by policy makers, and public–private partnerships promoted, along with flexible financing structures that help reduce the risk of investment. With techno-economic modeling and policy interventions applied strategically, RC technologies can establish themselves in the global PV market, driving cost reductions as well as energy transition to sustainable energy.

5.5. Field implementation and industry collaboration

Effective deployment of Radiative Cooling (RC) in Photovoltaics (PV) systems requires industry-initiated pilot projects and facilitating regulatory frameworks combined with private–public sector collaboration. While techno-economic models (Section 5.4) consider the cost viability and investment modalities for RC, this section considers real-world deployment experiences, rollout strategies, and market challenges that stimulate large-scale adoption. A structured approach integrating industry-led projects, public–private partnerships (PPPs), and regulatory inducements will be needed to bridge the gap between research advances and commercial deployment.

5.5.1. Industry-led pilot projects and large-scale deployment

Industry-led pilot projects have been instrumental in validating the performance and economic viability of RC-enhanced PV systems. Field deployments have demonstrated RC capability to boost PV efficiency, decrease operational expenses, and make the systems more sustainable. In China, field tests in low-latitude seaside regions showed that RC coatings reduced PV module temperatures by 6.5 °C, significantly improving efficiency under solar irradiance of 750 W/m^2 [189]. Also, studies in Pakistan found that the application of RC coatings on rooftops reflected yearly cooling load reduction values between 76.01–112.77 kWh/m^2, equivalent to electricity savings between 30.40–45.11 kWh/m^2 [49].

Further supporting these results, hybrid PV-RC systems in Las Vegas, USA, with heat pipe-thermoelectric generators reduced PV temperatures by up to 13 °C, boosting power output by 7.2 % [190]. These findings emphasize RC performance's climatic and geographical dependencies, and arid and semi-arid regions such as the Middle East, Australia, and India are most appropriate for deployment due to their high solar irradiance and low humidity [191]. Establishing large-scale industry testing facilities in collaboration with research institutions can further validate RC technologies under different environmental conditions, thereby ensuring long-term reliability and commercial feasibility.

5.5.2. Public-private partnerships (PPPs) for large-scale deployment

Public-private collaborations (PPPs) are most essential for closing gap between laboratory-scale research and commercial adoption. Governments may induce RC uptake through co-financing deployment initiatives, subsidizing setup expenses, and encouraging private funding. The Horizon 2020 Project under the EU financing revealed the viability of government-subsidized passive cooling technologies with 30 % energy-saving upgrade kits added on industrial cooling facilities [186]. The policymakers in Turkey have established renewable energy subsidies and feed-in tariffs (FITs) to promote energy-efficient cooling technologies, identifying these policy-created financial tools as motivators for the adoption of technology [192].

In the United States, California's Title 24 Energy Efficiency Standards have successfully mandated cool roofing materials, a policy framework that could be extended to RC-enhanced PV systems to encourage industry-wide adoption [193]. In addition, utility-backed deployment models, where electric utilities offer rebates and performance-based incentives for businesses that integrate RC-enhanced PV systems, could accelerate adoption and enhance grid stability [183]. For PPPs to be effective, governments must establish structured financial incentives, including direct subsidies, tax credits, and performance-linked tariffs. By supporting early adopters and industry-driven research collaborations, policymakers can lower financial risks, improve investor confidence, and facilitate large-scale RC-PV integration into national renewable energy policies.

5.5.3. Market barriers and regulatory challenges

Despite the proven benefits of RC technologies, several market and regulatory challenges hinder their widespread adoption. One major concern is high initial costs, particularly for the fabrication and large-scale production of high-performance RC materials. Even though the price has been decreasing gradually via innovation in nanophotonic and spectrally selective materials, producers are still not increasing production because of uncertainty about market demand and standardization [113].

Another significant hurdle is the lack of universal RC-enhanced PV material certification programs. Unlike conventional PV systems with matured efficiency ratings, RC materials have no standardized performance testing protocols, creating uncertainty within the investment community and tardy commercialization [187]. Governments and standards agencies must engage with industry players in formulating standard efficiency ratings and durability test protocols for guarantying reproducible performance testing across climatic regions.

Furthermore, geographical and climatic conditions have a significant influence on the variability of the performance of RC systems. Research shows that RC systems perform optimally in dry climates with high solar irradiance, while high humidity and cloud cover can reduce efficiency by absorbing infrared radiation, thus limiting their cooling potential [194]. This variability presents the necessity for region-tailored deployment strategies, with the priority implementation of RC in arid and semi-arid areas with abundant sunlight.

5.5.4. Recommendations for industry and policymakers

To address the obstacles preventing widespread implementation of Radiative Cooling (RC) technologies, industry leaders and policymakers need to adopt specific strategies that tackle cost issues, regulatory deficiencies, and technological constraints. One of the most effective ways to enhance market penetration is scaling up manufacturing capabilities. Lessons from the solar PV industry show that mass production can reduce costs by up to 40 %, making RC materials more affordable and accessible for widespread deployment [195]. Establishing high-volume manufacturing plants and encouraging technology transfer partnerships between research institutions and commercial producers will be essential to reducing production costs and improving material availability.

Concurrently, building regulatory support is most important to investor confidence and an explicitly defined sequence of adoption. Policymakers need to incorporate RC technologies into national energy efficiency codes, as California's Title 24 did with the use of high-reflectivity roof products in new construction to improve cooling performance [186]. Developing high-volume factory plants and technology transfer collaborations among research institutions and commercial producers will be crucial in lowering production cost and increasing material supply.

Additionally, developing hybrid cooling systems that integrate RC with thermoelectric or heat pipe cooling can enhance efficiency and minimize dependency on environmental conditions such as humidity and cloud cover. Research has proven that hybrid cooling designs hold

the possibility of increasing PV system efficiency by up to 7.2 %, particularly in arid and semi-arid regions where cooling demands are greatest [190]. It will be essential in realizing such hybrid designs to enable cross-industry collaboration between material scientists, PV manufacturers, and thermal engineers to optimize them for commercial adoption. By implementing these strategies, industry and policymakers can accelerate the transition of RC technologies from research labs to mainstream market adoption, ensuring cost efficiency, energy savings, and long-term sustainability in the global renewable energy sector.

6. Conclusion

This review demonstrates the significant potential of radiative cooling technologies in addressing thermal management challenges within Photovoltaic (PV) systems. Key findings reveal that advancements in coatings, photonic structures, and innovative materials have successfully improved PV module efficiency and durability by mitigating overheating and extending operational lifespans. The integration of passive cooling techniques, such as radiative cooling, into advanced PV configurations like bifacial and hybrid systems further underscores the promise of these technologies in enhancing energy efficiency and reducing the Levelized Cost Of Electricity (LCOE).

Despite these strides, significant challenges remain, including the scalability of fabrication processes, environmental durability of cooling materials, and the absence of comprehensive techno-economic evaluations. Furthermore, limited real-world testing under diverse environmental conditions highlights the need for region-specific solutions and performance validation. Addressing these gaps will require innovative approaches to material science, scalable manufacturing techniques, and robust frameworks for testing and evaluation.

Collaboration and interdisciplinary innovation are essential to overcome these challenges and unlock the full potential of radiative cooling for PV systems. By fostering partnerships among researchers, engineers, industry stakeholders, and policymakers, the field can develop scalable, cost-effective, and multifunctional solutions tailored to diverse climatic and operational requirements. AI-driven material optimization, integrated cooling strategies, and policy support can further accelerate the transition from experimental concepts to market-ready applications.

While radiative cooling has been very promising for PV performance improvement, there are a few areas that need research. The future effort should address improvement of material durability against UV radiation, humidity, and harsh environmental conditions. Manufacturing processes must be low cost and scalable to enable commercialization. Combining radiative cooling with new PV technologies such as tandem and perovskite solar cells offers a chance to get the maximum efficiency. Studies on hybrid cooling technologies, including the integration of radiative cooling with phase-change materials or active cooling, may further augment thermal management. Lastly, techno-economic analysis is essential in defining cost-effectiveness, energy saving, and market feasibility for mass application. Breakthroughs on these fronts will hasten the transition of radiative cooling from the laboratory to real-world, commercially viable applications for sustainable energy systems.

CRediT authorship contribution statement

Kehinde Temitope Alao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Syed Ihtsham Ul Haq Gilani:** Writing – review & editing, Supervision, Funding acquisition. **Kamaruzzaman Sopian:** Writing – review & editing, Supervision, Funding acquisition. **Taiwo Onaopemipo Alao:** Writing – review & editing. **Zeshan Aslam:** Writing – review & editing.

Acknowledgements

The authors of this work profoundly acknowledge the financial assistance received from PETRONAS – Malaysia via grant YUTP – FRG, Cost Center No: 015LC0-404.

References

[1] A. Razak, M. Yusoff, L. Wai Zhe, M. Irwanto, S. Ibrahim, M. Zhafarina, Investigation of the Effect Temperature on Photovoltaic (PV) panel output performance, Int J Adv Sci Eng Inf Technol 6 (2016) 682, https://doi.org/10.18517/ijaseit.6.5.938.

[2] D. Govindasamy, A. Kumar, Experimental analysis of solar panel efficiency improvement with composite phase change materials, Renew Energy 212 (2023) 175–184, https://doi.org/10.1016/j.renene.2023.05.028.

[3] T. Nabil, T.M. Mansour, Augmenting the performance of photovoltaic panel by decreasing its temperature using various cooling techniques, Results Eng. 15 (2022) 100564, https://doi.org/10.1016/j.rineng.2022.100564.

[4] R.E. Kassar, A.A. Takash, J. Faraj, M. Khaled, H.S. Ramadan, Phase change materials for enhanced photovoltaic panels performance: a comprehensive review and critical analysis, Energy Built Environ. (2024), https://doi.org/10.1016/j.enbenv.2024.02.004.

[5] K. Feng, Y. Wu, X. Pei, F. Zhou, Passive daytime radiative cooling: from mechanism to materials and applications, Mater Today Energy 43 (2024) 101575, https://doi.org/10.1016/j.mtener.2024.101575.

[6] S. Wijesuriya, R.A. Kishore, M.V.A. Bianchi, C. Booten, Potential energy savings benefits and limitations of radiative cooling coatings for U.S. residential buildings, J Clean Prod 379 (2022) 134763, https://doi.org/10.1016/j.jclepro.2022.134763.

[7] J. Wei, H. Chen, J. Liu, F. Wang, C. Wang, Radiative cooling technologies toward enhanced energy efficiency of solar cells: materials, systems, and perspectives, Nano Energy (2025), https://doi.org/10.1016/j.nanoen.2025.110680.

[8] H. Fang, L. Zhou, L. Xu, S. Dang, S. Wolf, Q. Gan, Radiative cooling for vertical solar panels, iScience 27 (2) (2024) 108806, https://doi.org/10.1016/j.isci.2024.108806.

[9] J. Liang, J. Wu, J. Guo, H. Li, X. Zhou, S. Liang, C.-W. Qiu, G. Tao, Radiative cooling for passive thermal management towards sustainable carbon neutrality, Natl Sci Rev 10 (2022), https://doi.org/10.1093/nsr/nwac208.

[10] X. Cui, X. Sun, L. Zhou, X. Wei, Thermal and electrical performance analysis of monofacial double-glass photovoltaic module with radiative cooling coating on the rear surface, Sol. Energy 264 (2023), https://doi.org/10.1016/j.solener.2023.111986.

[11] J. Mandal, Y. Fu, A.C. Overvig, M. Jia, K. Sun, N.N. Shi, H. Zhou, X. Xiao, N. Yu, Y. Yang, Hierarchically porous polymer coatings for highly efficient passive daytime radiative cooling, Science 1979 (362) (2018) 315–319, https://doi.org/10.1126/science:aat9513.

[12] A.P. Raman, M.A. Anoma, L. Zhu, E. Rephaeli, S. Fan, Passive radiative cooling below ambient air temperature under direct sunlight, Nature 515 (2014) 540–544, https://doi.org/10.1038/nature13883.

[13] Y. Zhai, Y. Ma, S.N. David, D. Zhao, R. Lou, G. Tan, R. Yang, X. Yin, Scalable-manufactured randomized glass-polymer hybrid metamaterial for daytime radiative cooling, Science 1979 (355) (2017) 1062–1066, https://doi.org/10.1126/science:aai7899.

[14] H. Fathabadi, Novel silica-based PV glass cover providing higher radiative cooling and power production compared with state-of-the-art glass covers, IEEE J Photovolt 11 (2021) 1485–1492, https://doi.org/10.1109/JPHOTOV.2021.3113160.

[15] N. Cheng, M. Dongyang, C. Wang, Y. Lin, A. Babar, X. Wang, Z. Wang, J. Yu, B. Ding, Nanosphere-structured hierarchically porous PVDF-HFP fabric for passive daytime radiative cooling via one-step water vapor-induced phase separation, Chem. Eng. J. 460 (2023) 141581, https://doi.org/10.1016/j.cej.2023.141581.

[16] J.-H. Fu, Q. Zhang, M. Yan, W. Tang, Y. Zeng, Y. Cui, M. Zhang, H. Deng, H. Kang, X. Sun, L. Qiao, H. Wang, Y. Zhao, A flexible and superhydrophobic PVDF-HFP@PVA bilayer enables high-performance radiative cooling and thermal management, Chem. Eng. J. 502 (2024) 157877, https://doi.org/10.1016/j.cej.2024.157877.

[17] Y. Liu, Y. Tian, X. Liu, F. Chen, A. Caratenuto, Y. Zheng, Intelligent regulation of VO 2 -PDMS-driven radiative cooling, Appl Phys Lett 120 (2022) 171704, https://doi.org/10.1063/5.0089353.

[18] M. Ghamari, C.H. See, D. Hughes, T. Mallick, K.S. Reddy, K. Patchigolla, S. Sundaram, Advancing sustainable building through passive cooling with phase change materials, a comprehensive literature review, Energy Build 312 (2024), https://doi.org/10.1016/j.enbuild.2024.114164.

[19] M. Hadjadj, T. Nehari, Z.A. Zehouani, B. Allam, Innovative passive techniques and PCM integration in building envelopes for enhanced energy efficiency in desert regions, J Energy Storage 103 (2024), https://doi.org/10.1016/j.est.2024.114322.

[20] O. McCay, R. Nimmagadda, S.M. Ali, T. Persoons, A parametric design study of natural-convection-cooled heat sinks, Fluids 8 (2023), https://doi.org/10.3390/fluids8080234.

[21] S. Pujari, P. Chintada, A.K. Inkulu, G.G. Deepa, Experimental investigations on heat sink circular pin fins and conical spines subjected to natural convection, Lecture Notes on Multidisciplinary Industrial Engineering (2019), https://doi.org/10.1007/978-981-13-7643-6_1.

[22] Bhatnagar, A., Hegde, A.K., H S, A., Karanth, K.V., N, M., 2024. Innovative cooling for PV panels: Energy and exergy assessments of water-induced V-shaped

channels. Results in Engineering 24, 103100. doi: 10.1016/j.rineng.2024.103100.
[23] A.K. Hamzat, A.Z. Sahin, M.I. Omisanya, L.M. Alhems, Advances in PV and PVT cooling technologies: a review, Sustainable Energy Technol. Assess. 47 (2021), https://doi.org/10.1016/j.seta.2021.101360.
[24] H.A. Hussien, A.H. Numan, A.R. Abdulmunem, Improving of the photovoltaic / thermal system performance using water cooling technique, in: In: IOP Conference Series: Materials Science and Engineering, 2015, https://doi.org/10.1088/1757-899X/78/1/012020.
[25] H. Sainthiya, N.S. Beniwal, Comparative analysis of electrical performance parameters under combined water cooling technique of photovoltaic module: an experimental investigation, Energy Sources Part A 42 (2020) 1902–1913, https://doi.org/10.1080/15567036.2019.1604894.
[26] A. Amr, A. Hassan, M. Abdel-Salam, A. El-Sayed, Active cooling of a photovoltaic module in hot-ambient temperatures: theory versus experiment, Int. J. Emerg. Electr. Power Syst. (2024), https://doi.org/10.1515/ijeeps-2023-0398.
[27] A.F. Eid, S.-I. Lee, S.-G. Hong, W. Choi, Hybrid cooling techniques to improve the performance of solar photovoltaic modules, Sol. Energy 245 (2022) 254–264, https://doi.org/10.1016/j.solener.2022.09.026.
[28] H. Metwally, N.A. Mahmoud, M. Ezzat, W. Aboelsoud, Numerical investigation of photovoltaic hybrid cooling system performance using the thermoelectric generator and RT25 Phase change material, J Energy Storage 42 (2021), https://doi.org/10.1016/j.est.2021.103031.
[29] H. Nasef, Integrative passive and active cooling system using PCM and nanofluid for thermal regulation of concentrated photovoltaic solar cells, Energy Convers Manag 199 (2019), https://doi.org/10.1016/j.enconman.2019.112065.
[30] Zhuang, X., Xu, X., Cui, J., 2019. Thermal management techniques for concentrating photovoltaic modules, Nanostructured Materials for Next-Generation Energy Storage and Conversion: Photovoltaic and Solar Energy. doi: 10.1007/978-3-662-59594-7_9.
[31] K. Amber, W. Akram, M.A. Bashir, M.S. Khan, A. Kousar, Experimental performance analysis of two different passive cooling techniques for solar photovoltaic installations, J Therm Anal Calorim 143 (2020), https://doi.org/10.1007/s10973-020-09883-6.
[32] S.S. Koohestani, S. Nižetić, M. Santamouris, Comparative review and evaluation of state-of-the-art photovoltaic cooling technologies, J Clean Prod 406 (2023), https://doi.org/10.1016/j.jclepro.2023.136953.
[33] S. Nižetić, E. Giama, A.M. Papadopoulos, Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels, Part II: active cooling techniques, Energy Convers Manag 155 (2018) 301–323, https://doi.org/10.1016/j.enconman.2017.10.071.
[34] A. Radwan, M. Emam, M. Ahmed, Chapter 2.15 - comparative study of active and passive cooling techniques for concentrated photovoltaic systems, in: I. Dincer, C. O. Colpan, O. Kizilkan (Eds.), Exergetic, Energetic and Environmental Dimensions, Academic Press, 2018, pp. 475–505.
[35] G. Rózsás, G. Bognár, G. Takács, B. Plesz, Optimized process for the manufacturing of integrated microchannel cooling devices in the back contact of concentrator solar cells, in: In: 2022 Symposium on Design, Test, Integration and Packaging of MEMS/MOEMS (DTIP), 2022, pp. 1–5, https://doi.org/10.1109/DTIP56576.2022.9911724.
[36] H. Fayaz, N.A. Rahim, M. Hasanuzzaman, A. Rivai, R. Nasrin, Numerical and outdoor real time experimental investigation of performance of PCM based PVT system, Sol. Energy 179 (2019) 135–150, https://doi.org/10.1016/j.solener.2018.12.057.
[37] F. Selimefendigil, D. Okulu, H.F. Oztop, Application of ternary nanofluid and rotating cylinders in the cooling system of photovoltaic/thermoelectric generator coupled module and computational cost reduction, Appl Therm Eng 250 (2024), https://doi.org/10.1016/j.applthermaleng.2024.123436.
[38] A. Akbar, G. Najafi, S. Gorjian, A. Kasaeian, M. Mazlan, Performance enhancement of a hybrid photovoltaic-thermal-thermoelectric (PVT-TE) module using nanofluid-based cooling: indoor experimental tests and multi-objective optimization, Sustainable Energy Technol. Assess. 46 (2021) 101276, https://doi.org/10.1016/j.seta.2021.101276.
[39] A. Bria, B. Raillani, D. Chaatouf, M. Salhi, S. Amraqui, A. Mezrhab, Effect of PCM thickness on the performance of the finned PV/PCM system, Mater Today Proc 72 (2023) 3617–3625, https://doi.org/10.1016/j.matpr.2022.08.409.
[40] A.K. Maryood, K.H. Hilal, S.A. Ghadhban, Effect of using hybrid cooling technique on the performance of PV panel, AIP Conf. Proc. 3105 (2024) 030008, https://doi.org/10.1063/5.0213119.
[41] S. Pervez, M. Kamran, S. udin, A. Ahad, M. Khan, M. Faiq, Development and performance analysis of hybrid photovoltaic/thermal (PV/T) system, Journal of Thermal Engineering 7 (Supp 14) (2021) 1936–1944, https://doi.org/10.18186/thermal.1051272.
[42] D. Singh, H. Chaubey, Y. Parvez, A. Monga, S. Srivastava, Performance improvement of solar PV module through hybrid cooling system with thermoelectric coolers and phase change material, Sol. Energy 241 (2022) 538–552, https://doi.org/10.1016/j.solener.2022.06.028.
[43] R. Daghigh, Y. Khaledian, Effective design, theoretical and experimental assessment of a solar thermoelectric cooling-heating system, Sol. Energy 162 (2018) 561–572, https://doi.org/10.1016/j.solener.2018.01.012.
[44] R. Gad, H. Mahmoud, S. Ookawara, H. Hassan, Evaluation of thermal management of photovoltaic solar cell via hybrid cooling system of phase change material inclusion hybrid nanoparticles coupled with flat heat pipe, J Energy Storage 57 (2023), https://doi.org/10.1016/j.est.2022.106185.
[45] V. Silva, J. Martinez, R. Heideier, J. Bernal, A. Gimenes, M. Udaeta, M. Saidel, A long-term analysis of the architecture and operation of water film cooling system for commercial pv modules, Energies (Basel) 14 (2021), https://doi.org/10.3390/en14061515.
[46] G. Perrakis, A.C. Tasolamprou, G. Kenanakis, E.N. Economou, S. Tzortzakis, M. Kafesaki, Passive radiative cooling and other photonic approaches for the temperature control of photovoltaics: a comparative study for crystalline silicon-based architectures, Opt Express 28 (2020) 18548–18565, https://doi.org/10.1364/OE.388208.
[47] I.M. Slauch, M.G. Deceglie, T.J. Silverman, V.E. Ferry, Model for characterization and optimization of spectrally selective structures to reduce the operating temperature and improve the energy yield of photovoltaic modules, ACS Appl Energy Mater 2 (2019) 3614–3623, https://doi.org/10.1021/acsaem.9b00347.
[48] A. Elsafi, M. Abdelrahim, M. Elgaili, K. Mroue, A. Samara, A. Zekri, G. Willers, K. Ilse, B. Aïssa, H. Qasem, M. Pasha, S.A. Mansour, A. Abdallah, Analyzing the effectiveness of various coatings to mitigate photovoltaic modules soiling in desert climate, Sol. Energy Mater. Sol. Cells 280 (2025) 113278, https://doi.org/10.1016/j.solmat.2024.113278.
[49] A.S. Farooq, X. Song, Z. Wang, P. Zhang, Techno-economic analysis on prospects and implementation of employing radiative cooling technology across broader atmospheric conditions, Energy Sustain. Dev. 76 (2023), https://doi.org/10.1016/j.esd.2023.101302.
[50] M.A. Kecebas, M.P. Menguc, A. Kosar, K. Sendur, Passive radiative cooling design with broadband optical thin-film filters, J Quant Spectrosc Radiat Transf 198 (2017) 1339–1351, https://doi.org/10.1016/j.jqsrt.2017.03.046.
[51] A. Syafiq, M.Z. Farah Khaleda, V. Balakrishnan, N.A. Rahim, Facile synthesis of hydrophobic thermal insulation Nano-TiO2/ZnO and SnO coating for solar cell, J. Nano Res. 80 (2023) 111–122, https://doi.org/10.4028/p-7FxTyg.
[52] M. Huang, X. Ren, G.H. Tang, Q. Sun, M. Du, Feasibility of realizing photothermal, photovoltaic, and radiative cooling with a flexible structure, Renew Energy 236 (2024), https://doi.org/10.1016/j.renene.2024.121364.
[53] H.S. Majdi, M.A. Mashkour, L.J. Habeeb, A.H. Sabry, Enhancement of energy transfer efficiency For Photovoltaic (Pv) systems by cooling the panel surfaces, Eastern-European Journal of Enterprise Technologies 4 (2021) 83–89, https://doi.org/10.15587/1729-4061.2021.238700.
[54] J. Dumoulin, E. Drouard, M. Amara, Radiative sky cooling of solar cells: fundamental modelling and cooling potential of single-junction devices, Sustain Energy Fuels 5 (2021) 2085–2096, https://doi.org/10.1039/d0se01536a.
[55] M.D. Kempe, Ultraviolet light test and evaluation methods for encapsulants of photovoltaic modules, Sol. Energy Mater. Sol. Cells 94 (2010) 246–253, https://doi.org/10.1016/j.solmat.2009.09.009.
[56] A. Alshammari, E. Almatrafi, M. Rady, Radiative coatings for solar cell cooling: materials, and applications, Sol. Energy 273 (2024) 112545, https://doi.org/10.1016/j.solener.2024.112545.
[57] Gok, A., Fagerholm, C.L., Gordon, D.A., Bruckman, L.S., French, R.H., 2015. Degradation of poly(ethylene-terephthalate) under accelerated weathering exposures. In: 2015 IEEE 42nd Photovoltaic Specialist Conference, PVSC 2015. doi: 10.1109/PVSC.2015.7355601.
[58] J. Wang, K. Li, J. Zhang, J. Feng, Transparent and superhydrophobic FHA/SiO<inf>2</inf> coatings with obvious anti-soiling performance for photovoltaic modules, Prog Org Coat 183 (2023), https://doi.org/10.1016/j.porgcoat.2023.107679.
[59] Sun, Y., Zhou, Z., Jin, X., Sun, X., Alam, M.A., Bermel, P., 2017. Radiative cooling for concentrating photovoltaic systems. In: Proceedings of SPIE - The International Society for Optical Engineering. doi: 10.1117/12.2273916.
[60] G.K. Dalapati, A.K. Kushwaha, M. Sharma, V. Suresh, S. Shannigrahi, S. Zhuk, S. Masudy-Panah, Transparent heat regulating (THR) materials and coatings for energy saving window applications: Impact of materials design, micro-structural, and interface quality on the THR performance, Prog Mater Sci 95 (2018) 42–131, https://doi.org/10.1016/j.pmatsci.2018.02.007.
[61] J. Seo, M. Choi, S. Yoon, B.J. Lee, Climate-dependent optimization of radiative cooling structures for year-round cold energy harvesting, Renew Energy 217 (2023), https://doi.org/10.1016/j.renene.2023.119166.
[62] Wang, W., Fernandez, N., Katipamula, S., 2017. Parametric analysis of a photonic radiative cooling system. In: ISES Solar World Congress 2017 - IEA SHC International Conference on Solar Heating and Cooling for Buildings and Industry 2017, Proceedings. pp. 1015–1025. doi: 10.18086/swc.2017.17.05.
[63] J. Guo, S. Ju, J. Shiomi, Design of a highly selective radiative cooling structure accelerated by materials informatics, Opt Lett 45 (2020) 343–346, https://doi.org/10.1364/OL.45.000343.
[64] Li, Z., Xie, L., 2017. Optimal design of photonic cooling film for buildings based on improved genetic algorithm. In: Advanced Engineering and Technology III - Proceedings of the 3rd Annual Congress on Advanced Engineering and Technology, CAET 2016. pp. 3–12. doi: 10.1201/9781315387222-2.
[65] J. He, Q. Zhang, Y. Zhou, Y. Chen, H. Ge, S. Tang, Bioinspired polymer films with surface ordered pyramid arrays and 3D hierarchical pores for enhanced passive radiative cooling, ACS Nano 18 (2024) 11120–11129, https://doi.org/10.1021/acsnano.3c12244.
[66] Y. An, C. Sheng, X. Li, Radiative cooling of solar cells: opto-electro-thermal physics and modeling, Nanoscale 11 (2019) 17073–17083, https://doi.org/10.1039/c9nr04110a.
[67] T.S. Balakrishnan, M.T.H. Sultan, F.S. Shahar, K.A. Ahmad, S.S. Dol, F.H. Naning, Bio-inspired coatings for natural fibre composites, J. Sci.: Adv. Mater. Devices 10 (2025) 100841, https://doi.org/10.1016/j.jsamd.2024.100841.
[68] A.A. Kirubaraj, D.J. Moni, D. Devaprakasam, Large scale fabrication of asymmetric 2D and 3D micro/nano array pattern structures using multi-beam interference lithography technique for Solar cell texturing application, Microsyst. Technol. 24 (2018) 2569–2575, https://doi.org/10.1007/s00542-018-3742-4.

[69] X. Wang, H. Lü, Q.-L. Zhao, S.-Y. Zhang, W.-Y. Tam, Application progress of holographic lithgraphy in fabrication of micro-nano photonic structures, Guang Pu Xue Yu Guang Pu Fen Xi/spectroscopy and Spectral Analysis 36 (2016) 3461–3469, https://doi.org/10.3964/j.issn.1000-0593(2016)11-3461-09.
[70] S. Haque, M. Alexandre, M.J. Mendes, H. Águas, E. Fortunato, R. Martins, Design of wave-optical structured substrates for ultra-thin perovskite solar cells, Appl Mater Today 20 (2020), https://doi.org/10.1016/j.apmt.2020.100720.
[71] H. Hanifi, M. Pander, U. Zeller, K. Ilse, D. Dassler, M. Mirza, M.A. Bahattab, B. Jaeckel, C. Hagendorf, M. Ebert, R. Gottschalg, J. Schneider, Loss analysis and optimization of PV module components and design to achieve higher energy yield and longer service life in desert regions, Appl Energy 280 (2020), https://doi.org/10.1016/j.apenergy.2020.116028.
[72] M. Mehdi, N. Ammari, A. Alami Merrouni, M. Dahmani, A. Benazzouz, Outdoor experimental investigation of the temperature effect on the performance of different PV modules materials, Key Eng. Mater. (2023), https://doi.org/10.4028/p-nqJi1m.
[73] K.-C. Kim, Hydrophobic sol-gel anti-reflective coating with various sized silica nano particles, Journal of Advanced Research in Dynamical and Control Systems 10 (2018) 201–206.
[74] K.-C. Kim, Hydrophobic sol-gel anti-reflective coating with various sized silica nanoparticles, J. Eng. Appl. Sci. 12 (2017) 8919–8923, https://doi.org/10.3923/jeasci.2017.8919.8923.
[75] S.J. Oh, S. Chhajed, D.J. Poxson, J. Cho, E.F. Schubert, S.J. Tark, D. Kim, J. K. Kim, Enhanced broadband and omni-directional performance of polycrystalline si solar cells by using discrete multilayer antireflection coatings, Opt Express 21 (2013) A157–A166, https://doi.org/10.1364/OE.21.00A157.
[76] G. Shang, Y. Häntsch, K.P. Furlan, R. Janßen, G.A. Schneider, A. Petrov, M. Eich, Highly selective photonic glass filter for saturated blue structural color, APL Photonics 4 (2019) 046101, https://doi.org/10.1063/1.5084138.
[77] S. Jin, S. Wang, H. Feng, H. He, M. Huang, S. Jin, D. Yang, L. Wang, Mechanically robust and self-cleaning antireflective coatings for photovoltaic modules, Sol. Energy Mater. Sol. Cells 275 (2024) 113009, https://doi.org/10.1016/j.solmat.2024.113009.
[78] T.-C. Chang, X. Cao, S.-H. Bao, S.-D. Ji, H.-J. Luo, P. Jin, Review on thermochromic vanadium dioxide based smart coatings: from lab to commercial application, Adv Manuf 6 (2018) 1–19, https://doi.org/10.1007/s40436-017-0209-2.
[79] I.G. Madiba, L. Kotsedi, B.D. Ngom, B.S. Khanyile, M. Maaza, Effect of substrate temperature on thermochromic vanadium dioxide thin films sputtered from vanadium target, AIP Conf. Proc. 10 (2018), https://doi.org/10.1063/1.5035540.
[80] X. Yi, Y. Li, S. Chen, S. Wang, J. Lai, C. Chen, H. Ma, Nanostructure and dropping phase transition temperature in vanadium dioxide thin films, Int J Nanosci 4 (2005) 99–106, https://doi.org/10.1142/S0219581X05002973.
[81] K.H. Jung, S.J. Yun, T. Slusar, H.-T. Kim, T.M. Roh, Highly transparent ultrathin vanadium dioxide films with temperature-dependent infrared reflectance for smart windows, Appl Surf Sci 589 (2022), https://doi.org/10.1016/j.apsusc.2022.152962.
[82] A. Sohail, M.S. Rusdi, M. Waseem, M.Z. Abdullah, F. Pallonetto, S.M. Sultan, Cutting-edge developments in active and passive photovoltaic cooling for reduced temperature operation, Results Eng. 23 (2024) 102662, https://doi.org/10.1016/j.rineng.2024.102662.
[83] Zhou, Z., Sun, X., Bermel, P., 2016. Radiative cooling for thermophotovoltaic systems. doi: 10.1117/12.2236174.
[84] Z. Zhou, Z. Wang, P. Bermel, Radiative cooling for low-bandgap photovoltaics under concentrated sunlight, Opt Express 27 (2019) A404–A418, https://doi.org/10.1364/OE.27.00A404.
[85] M. Hu, B. Zhao, X. Ao, J. Suhendri Cao, Q. Wang, S. Riffat, Y. Su, G. Pei, An analytical study of the nocturnal radiative cooling potential of typical photovoltaic/thermal module, Appl Energy 277 (2020), https://doi.org/10.1016/j.apenergy.2020.115625.
[86] S. Ahmed, Z. Li, M.S. Javed, T. Ma, A review on the integration of radiative cooling and solar energy harvesting, Mater Today Energy 21 (2021), https://doi.org/10.1016/j.mtener.2021.100776.
[87] Y. Zheng, Z. Fu, X. Sun, L. Zhou, X. Wei, J. Zhang, W. Xia, Y. Liu, Al<inf>2</inf>O<inf>3</inf>/graphene/PVDF-HFP radiative cooling coating reinforced heat dissipation of BIPV modules, J Appl Polym Sci 141 (2024), https://doi.org/10.1002/app.55451.
[88] L. Wei, N. Li, H. Liu, C. Sun, A. Chen, R. Yang, Y. Qin, H. Bao, Macro-nanoporous film with cauliflower-shaped fibers for highly efficient passive daytime radiative cooling, ACS Appl Mater Interfaces (2024), https://doi.org/10.1021/acsami.4c15294.
[89] D. Song, C. Xu, Y. Chen, J. He, Y. Zhao, P.-J. Li, W. Lin, F. Fu, Enhanced thermal and electrochemical properties of PVDF-HFP/PMMA polymer electrolyte by TiO2 nanoparticles, Solid State Ion 282 (2015) 31–36, https://doi.org/10.1016/j.ssi.2015.09.017.
[90] W. Hu, F. Zhang, X. Tan, Y. Tu, S. Nie, Antibacterial PVDF coral-like hierarchical structure composite film fabrication for self-cleaning and radiative cooling effect, ACS Appl Mater Interfaces 16 (2024) 19828–19837, https://doi.org/10.1021/acsami.4c01926.
[91] B. Li, B. Cao, R. Song, Y. Zhong, C. Chen, H. Hu, S. Wu, Z. Xia, Low-cost and scalable sub-Ambient radiative cooling porous films, J Photonics Energy 13 (2023), https://doi.org/10.1117/1.JPE.13.015501.
[92] J. Liu, Y. Zhou, Z. Zhou, Y. Du, C. Wang, X. Yang, Z. Lin, Z. Guo, J. Zhao, L. Ye, H. Zhang, J. Yan, Passive photovoltaic cooling: advances toward low-temperature operation, Adv Energy Mater 14 (2024), https://doi.org/10.1002/aenm.202302662.
[93] J. Zeng, H. Shen, K. Gao, Y. Xu, Y. Li, H. Li, L. Wang, Effect of a radiative cooling emitter with indium tin oxide near-infrared heat mirror on the performance of photovoltaic module, Energy Sources Part A 46 (2024) 4504–4513, https://doi.org/10.1080/15567036.2024.2329813.
[94] J. Shang, J. Zhang, Y. Zhang, X. Zhang, Q. An, Highly potent transparent passive cooling coating via microphase-separated hydrogel combining radiative and evaporative cooling, Nano Lett 24 (2024) 7055–7062, https://doi.org/10.1021/acs.nanolett.4c01621.
[95] Q. Dai, P. Xiang, B. Li, X. Yang, Z. Guo, H. Zhang, Q. Luo, C. Ju, H. Weng, Y. Liu, Y. Liu, X. Tan, Highly reliable anti-reflection radiative cooling glass applicable to thermal management of solar cells, Chem. Eng. J. 498 (2024), https://doi.org/10.1016/j.cej.2024.155768.
[96] J. Mao, X. Tan, W. Hu, C. Shi, F. Zhou, A. Tsamis, Simple preparation of PVDF composite flexible film with transparent, self-cleaning and radiative cooling properties, RSC Adv 14 (2024) 36656–36666, https://doi.org/10.1039/d4ra06819j.
[97] M. Hu, B. Zhao, X. Ao, J. Suhendri Cao, Q. Wang, S. Riffat, Y. Su, G. Pei, An analytical study of the nocturnal radiative cooling potential of typical photovoltaic/thermal modules, Appl Energy 277 (2020), https://doi.org/10.1016/j.apenergy.2020.115625.
[98] M. Hu, B. Zhao, X. Ao, J. Suhendri Cao, Q. Wang, S. Riffat, Y. Su, G. Pei, An analytical study of the nocturnal radiative cooling potential of typical photovoltaic/thermal modules, Appl Energy 277 (2020), https://doi.org/10.1016/j.apenergy.2020.115625.
[99] W. Pang, Y. Liu, S. Shao, X. Gao, Empirical study on thermal performance through separating impacts from a hybrid PV/TE system design integrating heat sink, Int. Commun. Heat Mass Transfer 60 (2015) 9–12, https://doi.org/10.1016/j.icheatmasstransfer.2014.11.004.
[100] B. Sun, L. Lu, J. Chen, T. Ma, Y. Yuan, Full-spectrum radiative cooling for enhanced thermal and electrical performance of bifacial solar photovoltaic modules: a nationwide quantitative analysis, Appl Energy 362 (2024), https://doi.org/10.1016/j.apenergy.2024.123037.
[101] M. Hu, B. Zhao, A.O. Suhendri, C. X, J. Cao, Q. Wang, S. Riffat, Y. Su, G. Pei, Applications of radiative sky cooling in solar energy systems: progress, challenges, and prospects, Renew. Sustain. Energy Rev. 160 (2022) 112304, https://doi.org/10.1016/j.rser.2022.112304.
[102] J. Jaramillo-Femandez, G. Silva-Oelker, D. Astrain, Passive radiative cooling materials integrated in renewable energy technologies for enhanced performance, EPJ Web Conf. 309 (2024) 13007, https://doi.org/10.1051/epjconf/202430913007.
[103] A. Bhardwaj, N.M. Puthoor, H.R. Matte, G.G. Nair, Metal-free radiative cooling polymer films containing high bandgap materials employing a tandem approach, J Quant Spectrosc Radiat Transf 298 (2023), https://doi.org/10.1016/j.jqsrt.2023.108495.
[104] L. Carlosena, Á. Andueza, L. Torres, O. Irulegi, R.J. Hernández-Minguillón, J. Sevilla, M. Santamouris, Experimental development and testing of low-cost scalable radiative cooling materials for building applications, Sol. Energy Mater. Sol. Cells 230 (2021), https://doi.org/10.1016/j.solmat.2021.111209.
[105] Xiao, C., Chen, C., 2024. Recycled Glass-Based Radiative Cooling Paint: A Feasible Approach for Space Cooling Applications, in: 18th Conference of the International Society of Indoor Air Quality and Climate, INDOOR AIR 2024 - Conference Program and Proceedings.
[106] S. Zhang, Z. Wu, Z. Liu, H. Lin, Z. Lin, J. Li, S. Kong, Z. Hu, Cost effective 24-h radiative cooler with multiphase interface enhanced solar scattering and thermal emission, Mater Today Commun 31 (2022), https://doi.org/10.1016/j.mtcomm.2022.103398.
[107] R. Kumar, F.J. Montero, R. Lamba, M. Vashishtha, S. Upadhyaya, Thermal management of photovoltaic-thermoelectric generator hybrid system using radiative cooling and heat pipe, Appl Therm Eng 227 (2023), https://doi.org/10.1016/j.applthermaleng.2023.120420.
[108] J. Niu, W. Zhou, J. Bai, S. Wang, X. Ma, S. Li, Y. Fan, A multi-scale particles doped ETFE based self-cleaning coating for passive daytime radiative cooling, Mater Today Commun 39 (2024), https://doi.org/10.1016/j.mtcomm.2024.108918.
[109] J. Song, Q. Shen, H. Shao, X. Deng, Anti-Environmental aging passive daytime radiative cooling, Adv. Sci. 11 (2024), https://doi.org/10.1002/advs.202305664.
[110] J.P. Bijarniya, J. Sarkar, P. Maiti, Environmental effect on the performance of passive daytime photonic radiative cooling and building energy-saving potential, J Clean Prod 274 (2020), https://doi.org/10.1016/j.jclepro.2020.123119.
[111] J. Huang, C. Lin, Y. Li, B. Huang, Effects of humidity, aerosol, and cloud on subambient radiative cooling, Int J Heat Mass Transf 186 (2022), https://doi.org/10.1016/j.ijheatmasstransfer.2021.122438.
[112] L. Carlosena, Á. Ruiz-Pardo, E.Á. Rodríguez-Jara, M. Santamouris, Worldwide potential of emissive materials based radiative cooling technologies to mitigate urban overheating, Build Environ 243 (2023), https://doi.org/10.1016/j.buildenv.2023.110694.
[113] J. Liu, Z. Zhou, J. Zhang, W. Feng, J. Zuo, Advances and challenges in commercializing radiative cooling, Mater. Today Phys. 11 (2019), https://doi.org/10.1016/j.mtphys.2019.100161.
[114] M.A. de Souza, J.P. de Souza, A.T. Pereima, Simplified methodology for temperature calculation of operation and photovoltaic modules yield in non-standardized environmental conditions, Braz. Arch. Biol. Technol. 61 (2018), https://doi.org/10.1590/1678-4324-smart-2018000170.
[115] A. Habte, A. Andreas, L. Ottoson, C.A. Gueymard, G. Fedor, S. Fowler, J. Peterson, E. Naranen, T. Kobashi, A. Akiyama, A. Akiyama, S. Takagi, Spectroradiometer intercomparison and impact on characterizing photovoltaic device performance, in: 43rd ASES National Solar Conference 2014, SOLAR 2014, including the 39th

National Passive Solar Conference and the 2nd Meeting of Young and Emerging Professionals in Renewable Energy, 2014, pp. 794–801.
[116] J. Campbell, Y. Zemen, B. Richardson, B. Striner, Photovoltaic module performance and degradation as compared in distinct climatic regions, Conf. Rec. IEEE Photovolt. Spec. Conf. (2012) 1250–1255, https://doi.org/10.1109/PVSC.2012.6317829.
[117] M.D. Kempe, D. Holsapple, K. Whitfield, N. Shiradkar, Standards development for modules in high temperature micro-environments, Prog. Photovolt. Res. Appl. 29 (2021) 445–460, https://doi.org/10.1002/pip.3389.
[118] D. Sankaran, B. Pappan, K. Balakrishnan, Enhancing photo voltaic solar panel efficiency by using a combination of silica based and phase change material coating, Mater Res Express 10 (2023), https://doi.org/10.1088/2053-1591/ad07cc.
[119] T. Xu, R. He, G. Fan, D. Zhang, J. Zhang, Y. Du, W. Zhou, J. Yang, Hexadecanol-palmitic acid/expanded graphite eutectic composite phase change material and its application in photovoltaic panel, Sol. Energy Mater. Sol. Cells 273 (2024), https://doi.org/10.1016/j.solmat.2024.112934.
[120] E. Sani, L. Mercatelli, D. Fontani, J.-L. Sans, D. Sciti, Hafnium and tantalum carbides for high temperature solar receivers, J. Renewable Sustainable Energy 3 (2011), https://doi.org/10.1063/1.3662099.
[121] Y. Liu, X. Bu, T. Yu, X. Wang, M. He, Z. Zhang, M. Feng, Y. Zhou, Design and scalable fabrication of core-shell nanospheres embedded spectrally selective single-layer coatings for durable daytime radiative cooling, Sol. Energy Mater. Sol. Cells 260 (2023), https://doi.org/10.1016/j.solmat.2023.112493.
[122] T. Rahman, A.A. Mansur, M.S. Hossain Lipu, M. Rahman, R.H. Ashique, M. Abou Houran, R. Madurai Elavarasan, E. Hossain, Investigation of degradation of solar photovoltaics: a review of aging factors, impacts, and future directions toward sustainable energy management, Energies (Basel) 16 (2023) 3706, https://doi.org/10.3390/en16093706.
[123] P.C. Okonkwo, S.C. Nwokolo, S.O. Udo, A.U. Obiwulu, U.N. Onnoghen, S. S. Alarifi, A.M. Eldosouky, S.E. Ekwok, P. Andráš, A.E. Akpan, Solar PV systems under weather extremes: case studies, classification, vulnerability assessment, and adaptation pathways, Energy Rep. 13 (2025) 929–959, https://doi.org/10.1016/j.egyr.2024.12.067.
[124] P. Yao, Z. Chen, T. Liu, X. Liao, Z. Yang, J. Li, Y. Jiang, N. Xu, W. Li, B. Zhu, B. Zhu, J. Zhu, Spider-Silk-inspired nanocomposite polymers for durable daytime radiative cooling, Adv. Mater. 34 (2022), https://doi.org/10.1002/adma.202208236.
[125] A. Zaite, N. Belouaggadia, C. Abid, A. Kaiss, H. Kanso, Integrate of night radiative cooling technology using a photovoltaic thermal collector under three different climates, International Journal of Thermofluids 16 (2022), https://doi.org/10.1016/j.ijft.2022.100252.
[126] I. Al-Masalha, A.S. Alsabagh, O. Badran, N. Alkawaldeh, T.M. Abu-Rahmeh, A. Al Alawin, Improving photovoltaic module efficiency using water sprinklers, air fans, and combined cooling systems, EPJ Photovoltaics 15 (2024), https://doi.org/10.1051/epjpv/2024037.
[127] Witmer, L., Brownson, J., 2011. An energy balance model of green roof integrated photovoltaics: A detailed energy balance including microclimatic effects. In: 40th ASES National Solar Conference 2011, SOLAR 2011. pp. 935–940.
[128] S.S. Bhakre, P.D. Sawarkar, V.R. Kalamkar, Performance evaluation of PV panel surfaces exposed to hydraulic cooling – a review, Sol. Energy 224 (2021) 1193–1209, https://doi.org/10.1016/j.solener.2021.06.083.
[129] Y. Zhou, T. Xia, X. Niu, S. Sun, L. Xu, Y. Jian, H. Wang, Enhancing radiative cooling performance for bifacial photovoltaic module using two kinds of polycarbonate films, J Photonics Energy 12 (2022), https://doi.org/10.1117/1.JPE.12.045501.
[130] T. Xia, H. Wang, High reflective polyethylene glycol terephthalate package layer for passive daytime radiative cooling in photovoltaic cells, Sol. Energy 237 (2022) 313–319, https://doi.org/10.1016/j.solener.2022.04.015.
[131] T. Ma, A. Kazemian, A. Habibollahzade, A. Salari, W. Gu, J. Peng, A comparative study on bifacial photovoltaic/thermal modules with various cooling methods, Energy Convers Manag 263 (2022), https://doi.org/10.1016/j.enconman.2022.115555.
[132] P. Govindasamy, B. Ramraj, G. Singaram, P. Sasirekha, V. Sabarivelu, S. Rajan, Electricity generation of dynamic bifacial solar panels using IoT, in: In: International Conference on Self Sustainable Artificial Intelligence Systems, 2023, pp. 1605–1610, https://doi.org/10.1109/ICSSAS57918.2023.10331709.
[133] A. Albatayneh, M. Jaradat, M. Al-Omary, M. Zaquot, Evaluation of coupling pv and air conditioning vs. Solar cooling systems—case study from jordan, Applied Sciences (Switzerland) 11 (2021) 1–15, https://doi.org/10.3390/app11020511.
[134] S. Nižetić, A.M. Papadopoulos, E. Giama, Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels, Part I: passive cooling techniques, Energy Convers Manag 149 (2017) 334–354, https://doi.org/10.1016/j.enconman.2017.07.022.
[135] S. Michas, V. Stavrakas, N.-A. Spyridaki, A. Flamos, Identifying Research Priorities for the further development and deployment of Solar Photovoltaics, Int. J. Sustain. Energ. 38 (2019) 276–296, https://doi.org/10.1080/14786451.2018.1495207.
[136] S. Ahmed, Z. Li, T. Ma, M.S. Javed, H. Yang, A comparative performance evaluation and sensitivity analysis of a photovoltaic-thermal system with radiative cooling, Sol. Energy Mater. Sol. Cells 221 (2021), https://doi.org/10.1016/j.solmat.2020.110861.
[137] Y. Cui, J. Zhu, S. Zoras, L. Liu, Review of the recent advances in dew point evaporative cooling technology: 3E (energy, economic and environmental) assessments, Renew. Sustain. Energy Rev. 148 (2021), https://doi.org/10.1016/j.rser.2021.111345.
[138] I. Marinić-Kragić, S. Nižetić, F. Grubišić-Čabo, D. Čoko, Analysis and optimization of passive cooling approach for free-standing photovoltaic panel: Introduction of slits, Energy Convers Manag 204 (2020), https://doi.org/10.1016/j.enconman.2019.112277.
[139] R. Braun, M. Haag, J. Stave, N. Abdelnour, U. Eicker, System design and feasibility of trigeneration systems with hybrid photovoltaic-thermal (PVT) collectors for zero energy office buildings in different climates, Sol. Energy 196 (2020) 39–48, https://doi.org/10.1016/j.solener.2019.12.005.
[140] Czapla, B., Srinivasan, A., Yin, Q., Narayanaswamy, A., 2017. Potential for passive radiative cooling by PDMS selective emitters, in: ASME 2017 Heat Transfer Summer Conference, HT 2017. doi: 10.1115/HT2017-4853.
[141] H. Ju, S. Lei, F. Wang, D. Yang, J. Ou, A. Amirfazli, Daytime radiative cooling performance and building energy consumption simulation of superhydrophobic calcined kaolin/poly(vinylidene fluoride-co-hexafluoropropylene) coatings, Energy Build 292 (2023), https://doi.org/10.1016/j.enbuild.2023.113184.
[142] Z. Wan, D. Liu, J. Yu, Research progress of daytime radiative cooling materials | 日间辐射制冷材料研究进展, Cailiao Daobao/materials Reports 36 (2022) https://doi.org/10.11896/cldb.20100091.
[143] X. Yin, R. Yang, G. Tan, S. Fan, Terrestrial radiative cooling: Using the cold universe as a renewable and sustainable energy source, Science 1979 (370) (2020) 786–791, https://doi.org/10.1126/science:abb0971.
[144] S. Nie, L. Bai, G. Lin, K. Yuan, J. Fu, Y. Zhang, H. Wang, H. Lan, P. Liu, X. Tan, X. Tan, X. Li, Experimental study on energy-free superhydrophobic radiative cooling versatile film with enhanced environmental tolerance, ACS Appl Mater Interfaces 16 (2024) 25498–25510, https://doi.org/10.1021/acsami.4c01209.
[145] E. Svinterikos, I. Zuburtikudis, H. Abu Khalifeh, S. Farvin Akbar Ali, Multifunctional polymer-based coatings for outdoor glass surfaces: a state of the art, Adv. Ind. Eng. Polym. Res. 6 (2023) 310–332, https://doi.org/10.1016/j.aiepr.2023.04.001.
[146] J. Liang, J. Wu, J. Guo, H. Li, X. Zhou, S. Liang, C.-W. Qiu, G. Tao, Radiative cooling for passive thermal management towards sustainable carbon neutrality, Natl Sci Rev 10 (2023), https://doi.org/10.1093/nsr/nwac208.
[147] Cagnoni, M., Tibaldi, A., Dolado, J.S., Cappelluti, F., 2022. Cementitious materials as promising radiative coolers for solar cells. iScience 25. doi: 10.1016/j.isci.2022.105320.
[148] S.-J. Park, S.-B. Seo, J. Shim, S. Jin Hong, G. Kang, H. Ko, S. Jeong, S.-K. Kim, Three-dimensionally printable hollow silica nanoparticles for subambient passive cooling, Nanophotonics 13 (2024) 611–620, https://doi.org/10.1515/nanoph-2023-0603.
[149] S. Zhang, W. Jing, Z. Chen, C. Zhang, D. Wu, Y. Gao, H. Zhu, Full daytime sub-ambient radiative cooling film with high efficiency and low cost, Renew Energy 194 (2022) 850–857, https://doi.org/10.1016/j.renene.2022.05.151.
[150] G. Perrakis, A.C. Tasolamprou, G. Kenanakis, E.N. Economou, S. Tzortzakis, M. Kafesaki, Passive radiative cooling for the temperature and efficiency control of photovoltaics. In: 2021 Conference on Lasers and Electro-Optics, 2021.
[151] H. Li, J. Zhang, X. Liu, T. Zhang, Comparative investigation of energy-saving potential and technical economy of rooftop radiative cooling and photovoltaic systems, Appl Energy 328 (2022), https://doi.org/10.1016/j.apenergy.2022.120181.
[152] P.M. Rodrigo, A. Valera, E.F. Fernández, F.M. Almonacid, Performance and economic limits of passively cooled hybrid thermoelectric generator-concentrator photovoltaic modules, Appl Energy 238 (2019) 1150–1162, https://doi.org/10.1016/j.apenergy.2019.01.132.
[153] A.S. Baharom, N.Y. Dahlan, Techno-economic influence on Malaysia's solar thermal installation for heating process, Indonesian Journal of Electrical Engineering and Computer Science 12 (2018) 600–606, https://doi.org/10.11591/ijeecs.v12.i2.pp600-606.
[154] A.D. Mills, R.H. Wiser, Changes in the economic value of photovoltaic generation at high penetration levels: a pilot case study of California, IEEE J Photovolt 3 (2013) 1394–1402, https://doi.org/10.1109/JPHOTOV.2013.2263984.
[155] P. Testa, M. Cagnoni, F. Cappelluti, Detailed-balance assessment of radiative cooling for multi-junction solar cells under unconcentrated and low-concentrated light, Sol. Energy Mater. Sol. Cells 274 (2024), https://doi.org/10.1016/j.solmat.2024.112958.
[156] Z. Wang, D. Kortge, J. Zhu, Z. Zhou, H. Torsina, C. Lee, P. Bermel, Lightweight, passive radiative cooling to enhance concentrating photovoltaics, Joule 4 (2020) 2702–2717, https://doi.org/10.1016/j.joule.2020.10.004.
[157] R. Lamba, F.J. Montero, T.-U. Rehman, S. Singh, S. Manikandan, PCM-based hybrid thermal management system for photovoltaic modules: a comparative analysis, Environ. Sci. Pollut. Res. 31 (2024) 46397–46416, https://doi.org/10.1007/s11356-023-27809-1.
[158] S. Al Arni, J.M. Mahdi, A.M. Abed, K.A. Hammoodi, H.A. Hasan, R.Z. Homod, N. B. Khedher, Novel multi-layer nano-modified PCM configuration for efficient thermal management of photovoltaic-thermal systems, J Energy Storage 103 (2024) 114352, https://doi.org/10.1016/j.est.2024.114352.
[159] Z. Khalili, M. Sheikholeslami, Investigation of innovative cooling system for photovoltaic solar unit in existence of thermoelectric layer utilizing hybrid nanomaterial and Y-shaped fins, Sustain Cities Soc 93 (2023) 104543, https://doi.org/10.1016/j.scs.2023.104543.
[160] M.S.F.M. Azmi, M.H. Hussain, S.R.A. Rahim, E.C. Mid, A.S. Shaari, N. Hashim, N. Husny, M.F. Ahmad, Hybrid cooling system for solar photovoltaic panel, J Phys Conf Ser 2550 (2023) 012004, https://doi.org/10.1088/1742-6596/2550/1/012004.
[161] R. Ismaen, M. Kucukvar, T.Y. El Mekkawy, A. Elomri, Optimization and enviro-economic assessment of solar-cooling systems towards sustainable development: a

case study of Qatar, J Clean Prod 419 (2023) 138253, https://doi.org/10.1016/j.jclepro.2023.138253.
[162] H. Carvalho, A. Raposo, R. Ines, P. Peças, A. Silva, E. Henriques, Influence of the use of natural fibers in composite materials assessed on a life cycle perspective, Handbook of Composites from Renewable Materials. (2017) 377–398, https://doi.org/10.1002/9781119441632.ch118.
[163] Q. Xuan, N. Yang, M. Kai, C. Wang, B. Jiang, X. Liu, G. Li, G. Pei, B. Zhao, Combined daytime radiative cooling and solar photovoltaic/thermal hybrid system for year-round energy saving in buildings, Energy 304 (2024) 132178, https://doi.org/10.1016/j.energy.2024.132178.
[164] F.C. Nkado, I.E. Aigwi, D.T. Doan, A. GhaffarianHoseini, Powering the future: a comprehensive analysis of solar photovoltaic systems adoption in passive houses, Sustainable Energy Technol. Assess. 73 (2025) 104166, https://doi.org/10.1016/j.seta.2024.104166.
[165] C. Yuan, Y. Shi, Z. Ba, D. Liang, J. Wang, X. Liu, Y. Xu, J. Liu, H. Xu, Machine learning models for predicting thermal properties of radiative cooling aerogels, Gels 11 (2025), https://doi.org/10.3390/gels11010070.
[166] C. Zhu, E.A. Bamidele, X. Shen, G. Zhu, B. Li, Machine learning aided design and optimization of thermal metamaterials, Chem Rev 124 (2024) 4258–4331, https://doi.org/10.1021/acs.chemrev.3c00708.
[167] Jung, Y., Ko, S.H., 2024. Radiative cooling technology with artificial intelligence. iScience 27. doi: 10.1016/j.isci.2024.111325.
[168] A. Sohani, H. Sayyaadi, C. Cornaro, M.H. Shahverdian, M. Pierro, D. Moser, N. Karimi, M.H. Doranehgard, L.K.B. Li, Using machine learning in photovoltaics to create smarter and cleaner energy generation systems: a comprehensive review, J Clean Prod 364 (2022), https://doi.org/10.1016/j.jclepro.2022.132701.
[169] M.J. Yoo, K.R. Pyun, Y. Jung, M. Lee, J. Lee, S.H. Ko, Switchable radiative cooling and solar heating for sustainable thermal management, Nanophotonics 13 (2024) 543–561, https://doi.org/10.1515/nanoph-2023-0627.
[170] C. Cai, X. Wu, F. Cheng, C. Ding, Z. Wei, X. Wang, Y. Fu, Cellulose metamaterials with hetero-profiled topology via structure rearrangement during ball milling for daytime radiative cooling, Adv Funct Mater 34 (2024) 2405903, https://doi.org/10.1002/adfm.202405903.
[171] Zhang, D., Wu, G., Du, F., Ji, Y., 2024. Enhancing photovoltaic efficiency with self-cleaning titanium dioxide nanoparticle coatings: a sustainability approach. In: Tenth International Conference on Energy Materials and Electrical Engineering (ICEMEE 2024). SPIE, pp. 150–155.
[172] D. Du, F. Wang, D. Zhang, J. Bao, Y. Fan, Y. Guo, W. Shen, H. Wang, One-step synthesis of SiO2 nanomesh for antireflection and self-cleaning of solar cell, J Colloid Interface Sci 630 (2023) 795–803, https://doi.org/10.1016/j.jcis.2022.10.157.
[173] M.S. Mohtasim, B.K. Das, MXene based composite phase change materials for thermal energy storage applications: featuring bio-mimic approaches, Renew. Sustain. Energy Rev. 207 (2025) 114952, https://doi.org/10.1016/j.rser.2024.114952.
[174] M. Yuan, C. Liu, C. Xu, Z. Liao, F. Ye, S. Liu, Dynamic absorption of bulk phase-change materials for photothermal solar energy storage based on reversible thermochromic | 基于可逆热致变色的动态体吸收太阳能光热存储相变材料, Zhongguo Kexue Jishu Kexue/scientia Sinica Technologica 53 (2023) 341–352, https://doi.org/10.1360/SST-2022-0359.
[175] D. Osuna Ruiz, C. Lezaun, A. Torres, M. Beruete, Metal-free design of a multilayered metamaterial with chirped Bragg grating for enhanced radiative cooling, Opt Express 31 (2023), https://doi.org/10.1364/OE.492404.
[176] Honey, S., Saidani-Scott, H., Coopey, H., Arkell, T., Cooke-Flores, A., Abdelhamid, A., 2023. Feasibility of Using Phase Change Material Calcium Chloride Hexahydrate to Improve the Efficiency of Bifacial Photovoltaic Modules via Passive Cooling. doi: 10.1109/ETFG55873.2023.10407975.
[177] M. Zahir, J. Drevillon, Y. Ezzahri, Improved thermophotovoltaic system performance with a novel metamaterial thermal emitter combined with a passive radiative cooling approach, Sol. Energy Mater. Sol. Cells 263 (2023) 112587, https://doi.org/10.1016/j.solmat.2023.112587.
[178] L. Mo, L. Yang, E.H. Lee, S. He, High-efficiency plasmonic metamaterial selective emitter based on an optimized spherical core-shell nanostructure for planar solar thermophotovoltaics, Plasmonics 10 (2015) 529–538, https://doi.org/10.1007/s11468-014-9837-6.
[179] W. Zou, J. Mo, Efficient heat dissipation of photovoltaic panels through in-place porous hydrogels for moisture harvesting and evaporation. Healthy Buildings 2023, Asia and Pacific Rim, 2023.
[180] R. Abu Taha, T. Daim, Multi-Criteria applications in renewable energy analysis, a literature review, Research and Technology Management in the Electricity Industry 60 (2013) 17–30, https://doi.org/10.1007/978-1-4471-5097-8_2.
[181] J. Peoples, Y.-W. Hung, Z. Fang, J. Braun, W.T. Horton, X. Ruan, Energy savings of radiative cooling paints applied to residential buildings, Int J Heat Mass Transf 194 (2022) 123001, https://doi.org/10.1016/j.ijheatmasstransfer.2022.123001.
[182] M. Sharaf, M.S. Yousef, A.S. Huzayyin, Review of cooling techniques used to enhance the efficiency of photovoltaic power systems, Environ. Sci. Pollut. Res. (2022), https://doi.org/10.1007/s11356-022-18719-9.
[183] Gomez, J., 2015. To own or lease solar installations: Considerations for decision makers, To Own or Lease Solar Installations: Considerations for Decision Makers.
[184] C. Muchunku, K. Ulsrud, D. Palit, W. Jonker Klunne, Diffusion of solar PV in East Africa: what can be learned from private sector delivery models? Wiley Interdiscip Rev Energy Environ 7 (2018) e282.
[185] H. BERTON, D. Rosenbloom, O. TEMBY, J. Meadowcroft, Using local improvement charges to finance solar photovoltaic systems, Energy Studies Review 22 (2015) 95–110, https://doi.org/10.15173/esr.v22i1.3308.
[186] C.B. Casady, A. Cepparulo, L. Giuriato, Public-private partnerships for low-carbon, climate-resilient infrastructure: insights from the literature, J Clean Prod 470 (2024) 143338, https://doi.org/10.1016/j.jclepro.2024.143338.
[187] H. Zhong, P. Zhang, Y. Li, X. Yang, Y. Zhao, Z. Wang, Highly Solar-Reflective structures for daytime radiative cooling under high humidity, ACS Appl Mater Interfaces 12 (2020) 51409–51417, https://doi.org/10.1021/acsami.0c14075.
[188] D. Han, J. Fei, H. Li, B.F. Ng, The criteria to achieving sub-ambient radiative cooling and its limits in tropical daytime, Build Environ 221 (2022) 109281, https://doi.org/10.1016/j.buildenv.2022.109281.
[189] J. Liu, Z. Zhou, D. Zhang, S. Jiao, Y. Zhang, L. Luo, Z. Zhang, F. Gao, Field investigation and performance evaluation of sub-ambient radiative cooling in low latitude seaside, Renew Energy 155 (2020) 90–99, https://doi.org/10.1016/j.renene.2020.03.136.
[190] R. Kumar, F.J. Montero, T.-U. Rehman, R. Lamba, M. Vashishtha, S. Upadhyaya, Radiative cooling system integrated with heat sink for the thermal management of photovoltaic modules under extreme climate conditions, J Therm Anal Calorim 148 (2023) 9099–9112, https://doi.org/10.1007/s10973-023-12291-1.
[191] R. Senthil Kumar, S. Manikandan, Mapping the all-day radiative sky-cooling potential of India for a sustainable environment, Energy Convers. Manage.: X 22 (2024), https://doi.org/10.1016/j.ecmx.2024.100591.
[192] M. Tükenmez, E. Demireli, Renewable energy policy in Turkey with the new legal regulations, Renew Energy 39 (2012) 1–9, https://doi.org/10.1016/j.renene.2011.07.047.
[193] Chakraborty, D., 2012. Advent of residential solar PV financing: Opportunities and challenges for the US installation industry and emerging trends. In: World Renewable Energy Forum, WREF 2012, Including World Renewable Energy Congress XII and Colorado Renewable Energy Society (CRES) Annual Conferen. pp. 2822–2826.
[194] Hwang, J., 2024. Daytime Radiative Cooling near the Equator, in: Specialty Optical Fibers, SOF 2024 in Proceedings Advanced Photonics Congress 2024 - Part of Optica Advanced Photonics Congress.
[195] S. So, J. Yun, B. Ko, D. Lee, M. Kim, J. Noh, C. Park, J. Park, J. Rho, Radiative cooling for energy sustainability: from fundamentals to fabrication methods toward commercialization, Adv. Sci. 11 (2024), https://doi.org/10.1002/advs.202305067.
[196] L. Li, Q. Zhang, G. Liu, R. Shi, H. Zhao, K. Huang, Y. Zang, Y. Xu, C. Li, L. Wu, Y. Zhang, Advanced passive daytime radiative cooling: from material selection and structural design to application towards multifunctional integration, Adv Compos Hybrid Mater 8 (2024) 97, https://doi.org/10.1007/s42114-024-01127-7.
[197] S. Ahmed, S. Li, Z. Li, G. Xiao, T. Ma, Enhanced radiative cooling of solar cells by integration with heat pipe, Appl Energy 308 (2022) 118363, https://doi.org/10.1016/j.apenergy.2021.118363.
[198] X. Cui, W. Ma, Y. Zheng, M. Luo, S. Han, X. Sun, L. Zhou, X. Wei, Thermal analysis of radiative cooling coating on the rear surface of photovoltaic tile, Sol. Energy 248 (2022) 210–220, https://doi.org/10.1016/j.solener.2022.11.007.
[199] K.W. Lee, W. Lim, M.S. Jeon, H. Jang, J. Hwang, C.H. Lee, D.R. Kim, Visibly clear radiative cooling metamaterials for enhanced thermal management in solar cells and windows, Adv Funct Mater 32 (2022) 2105882, https://doi.org/10.1002/adfm.202105882.
[200] Y. Lu, Z. Chen, L. Ai, X. Zhang, J. Zhang, J. Li, W. Wang, R. Tan, N. Dai, W. Song, A universal route to realize radiative cooling and light management in photovoltaic modules, Sol. RRL 1 (2017) 1700084, https://doi.org/10.1002/solr.201700084.
[201] Ahmed, S., Ma, T., 2020a. Performance comparison of different emissive layers in a combined photovoltaic-thermal and radiative cooling system, in: Energy Proceedings. doi: 10.46855/energy-proceedings-7494.
[202] Ahmed, S., Ma, T., 2020b. Performance comparison of different emissive layers in a combined photovoltaic-thermal and radiative cooling system, in: Energy Proceedings. doi: 10.46855/energy-proceedings-7494.
[203] Ahmed, S., Ma, T., 2020c. Performance comparison of different emissive layers in a combined photovoltaic-thermal and radiative cooling system, in: Energy Proceedings. doi: 10.46855/energy-proceedings-7494.
[204] A.A. Alshammari, E.M. Salilih, E. Almatrafi, M. Rady, Polymeric coatings for passive radiative cooling of PV modules in hot and humid weather: design, optimization, and performance evaluation, Case Stud. Therm. Eng. 57 (2024), https://doi.org/10.1016/j.csite.2024.104341.
[205] A. Basem, A. Mukhtar, Z.M.S. Elbarbary, F. Atamurotov, N.E. Benti, Experimental study on the various varieties of photovoltaic panels (PVs) cooling systems to increase their electrical efficiency, PLoS One 19 (2024), https://doi.org/10.1371/journal.pone.0307616.
[206] J. Chen, Q. Gong, L. Lu, Evaluation of passive envelope systems with radiative sky cooling and thermally insulated glazing materials for cooling, J Clean Prod 398 (2023), https://doi.org/10.1016/j.jclepro.2023.136607.
[207] Y.-H. Chen, C.-W. Hwang, S.-W. Chang, M.-T. Tsai, K.N. Jayakumaran, L.-C. Yang, Y.-C. Lo, F.-H. Ko, H.-C. Wang, H.-L. Chen, H.-L. Chen, D. Wan, Eco-Friendly transparent silk fibroin radiative cooling film for thermal management of optoelectronics, Adv Funct Mater 33 (2023), https://doi.org/10.1002/adfm.202301924.
[208] D. De giovanni, I. Elena, Renewable energy investments, support schemes and the dirty option, Appl Stoch Models Bus Ind 40 (2024), https://doi.org/10.1002/asmb.2901.
[209] Farahat, M.A., ElNaggar, S., Kader, A.M., Ahmed, S., Rashad, M.I., 2023. EXPERIMENTAL INVESTIGATION OF RADIATIVE COOLING POTENTIAL IN MEDITERRANEAN BASIN COUNTRIES, in: ASME International Mechanical

Engineering Congress and Exposition, Proceedings (IMECE). doi: 10.1115/IMECE2023-113012.
[210] M. Huang, H. Zhou, G.H. Tang, M. Du, Q. Sun, The potential of radiative cooling enhanced photovoltaic systems in China, Adv. Appl. Energy 15 (2024), https://doi.org/10.1016/j.adapen.2024.100184.
[211] Z. Li, S. Ahmed, T. Ma, Investigating the effect of radiative cooling on the operating temperature of photovoltaic modules, Sol. RRL 5 (2021), https://doi.org/10.1002/solr.202000735.
[212] H. Tang, W. Chen, C. Li, Experimental and theoretical study on building integrated radiative cooling with a limited sky-view factor, Journal of Building Engineering 97 (2024), https://doi.org/10.1016/j.jobe.2024.110948.
[213] W. Wang, H. Yang, C. Xiang, The overall performance of a novel semi-transparent photovoltaic window with passive radiative cooling coating – a comparative study, Energy Build 317 (2024), https://doi.org/10.1016/j.enbuild.2024.114433.
[214] X. Wang, S. Chen, L. Chen, D. Zhu, Y. Liu, T. Wu, A hybrid MLP-CNN model based on positional encoding for daytime radiative cooler, Opt Commun 560 (2024), https://doi.org/10.1016/j.optcom.2024.130448.
[215] B. Zhao, K. Lu, M. Hu, J. Liu, L. Wu, C. Xu, Q. Xuan, G. Pei, Radiative cooling of solar cells with micro-grating photonic cooler, Renew Energy 191 (2022) 662–668, https://doi.org/10.1016/j.renene.2022.04.063.
[216] Y. Zhou, H. Zhu, B. Chen, J. Chen, Y. Zhan, X. Li, Photonic radiative cooling power measurement system based on fuzzy control | 基于模糊控制的光子辐射制冷功率测量系统, Guangxue Xuebao/acta Optica Sinica 44 (2024) https://doi.org/10.3788/AOS231087.